Introduction and intertemporal choices

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UNIVERSITY OF COPENHAGEN

Course admin

Lectures every week

- Thursdays 1000h-1300h
- Office hours on lecture days from 1400h-1500h

Problem sets

- Exercise classes every week
- Preparation for these will greatly help with the exam

Assignments

- Pass 2/3 assignments to take the exam

Exam

- Closed book, 3h

What is macroeconomics?

Aggregate quantities

- GDP (growth & fluctuations)
- Inflation
- Unemployment
- ...

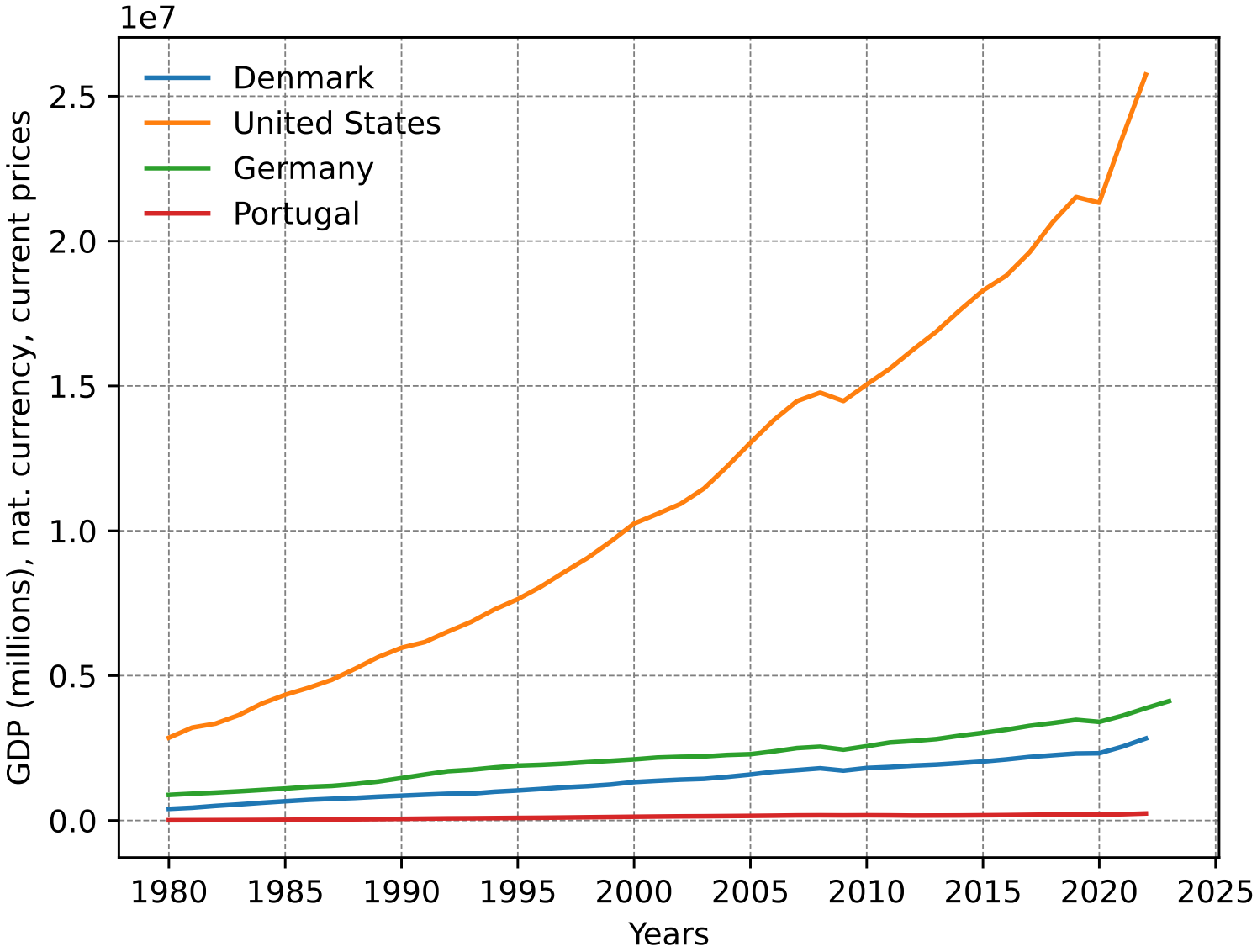
Society-wide phenomena

- Inequality
- Political changes

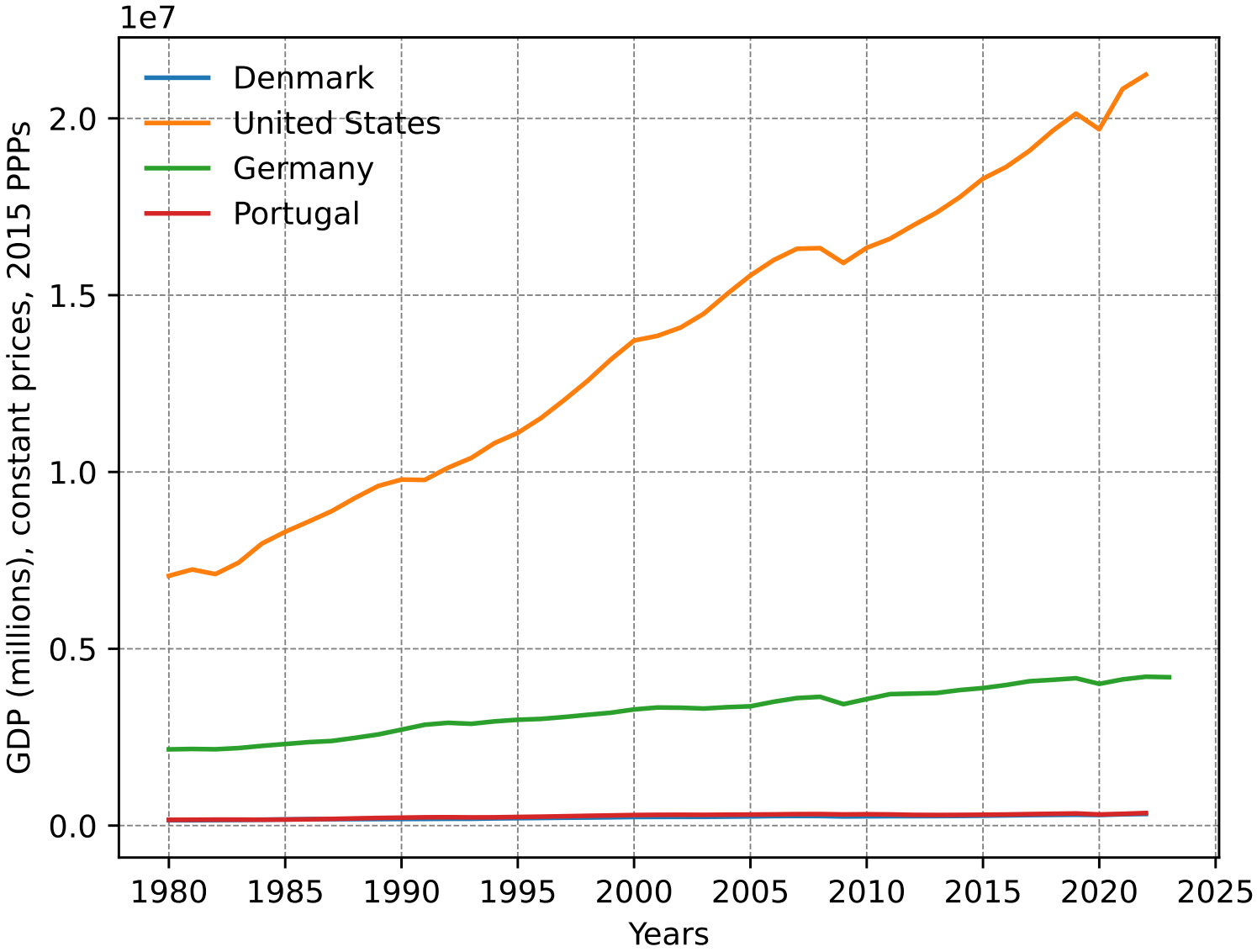
Public policy

- Pension systems
- Unemployment insurance

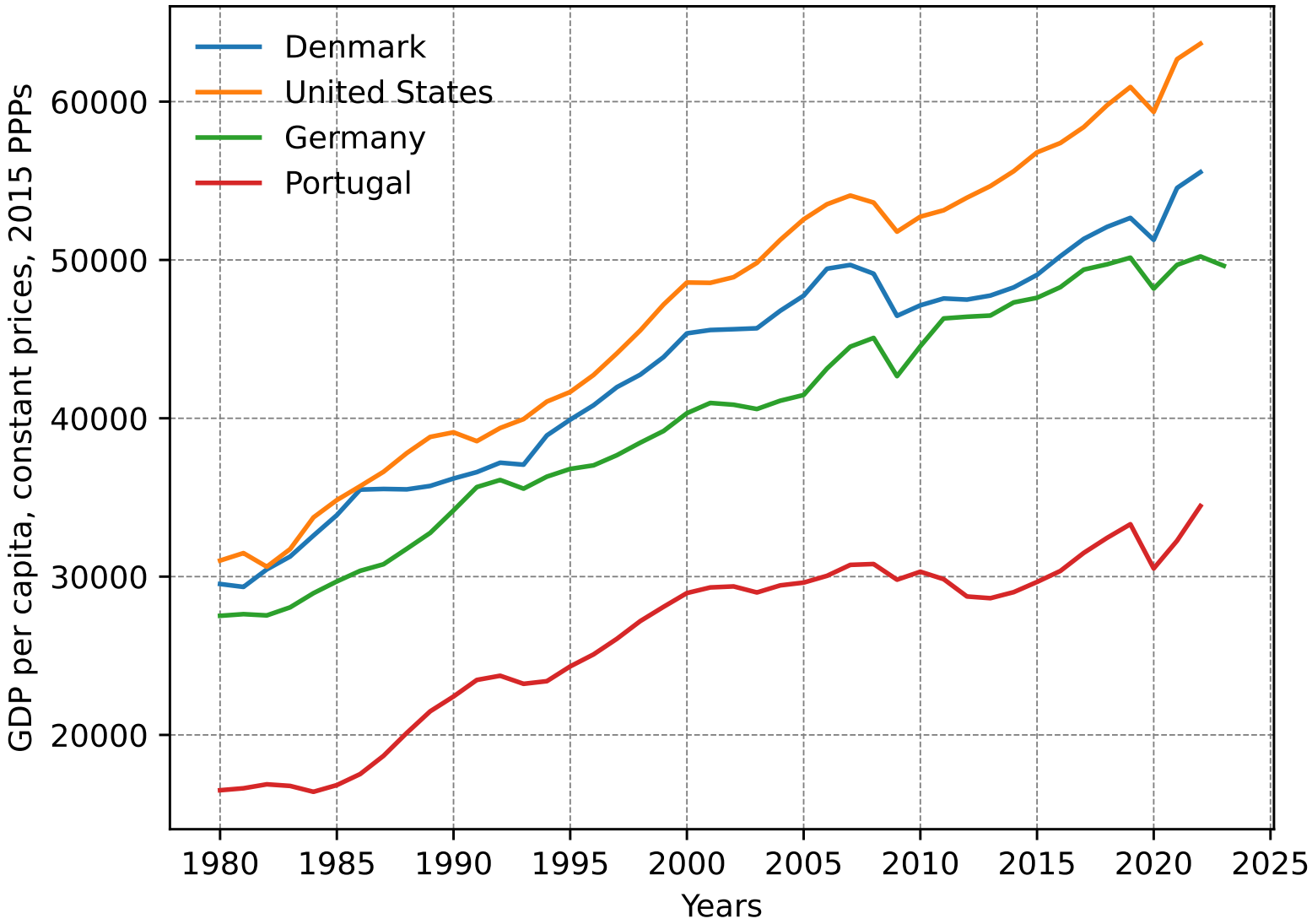
GDP across countries – OECD



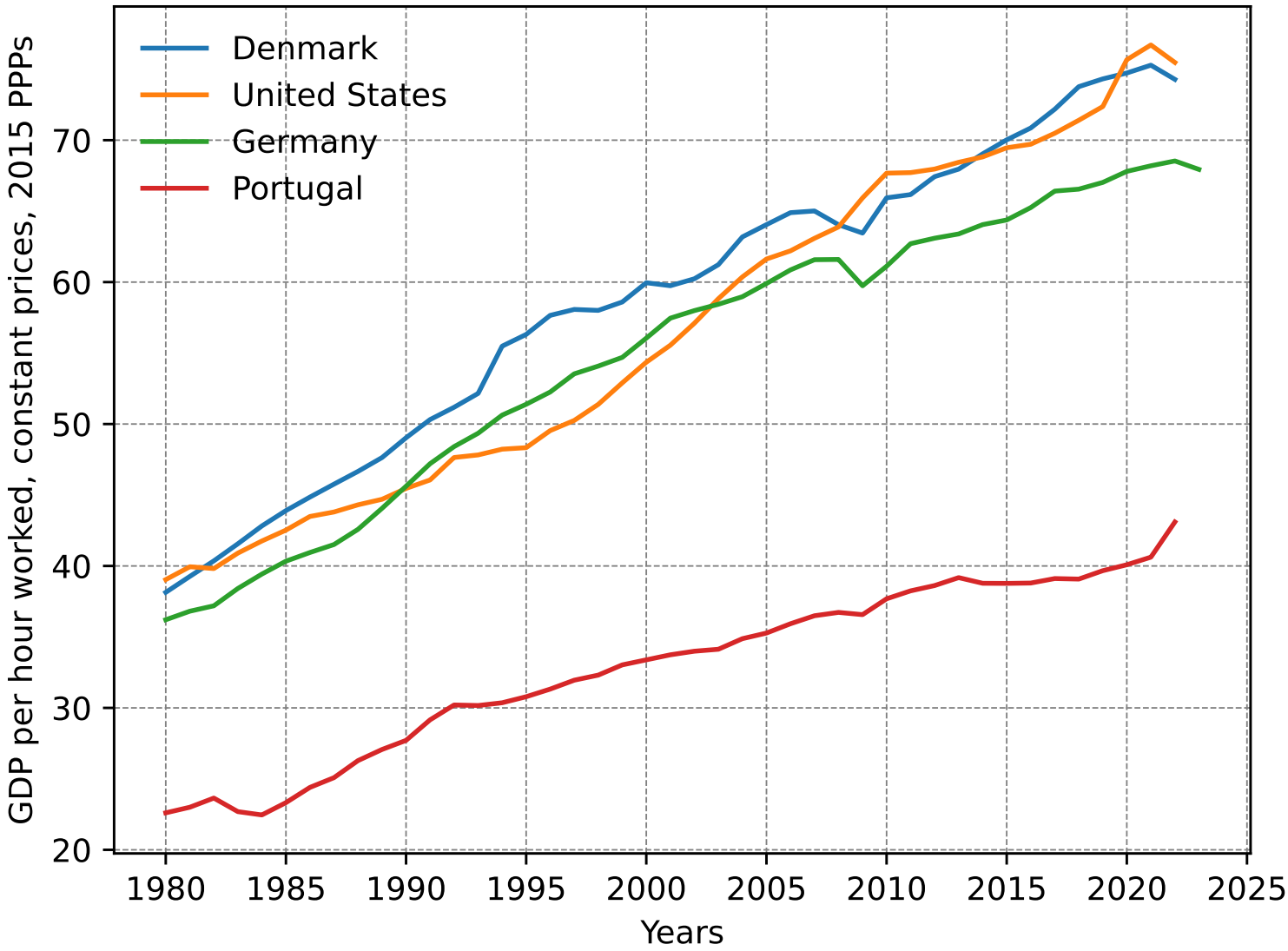
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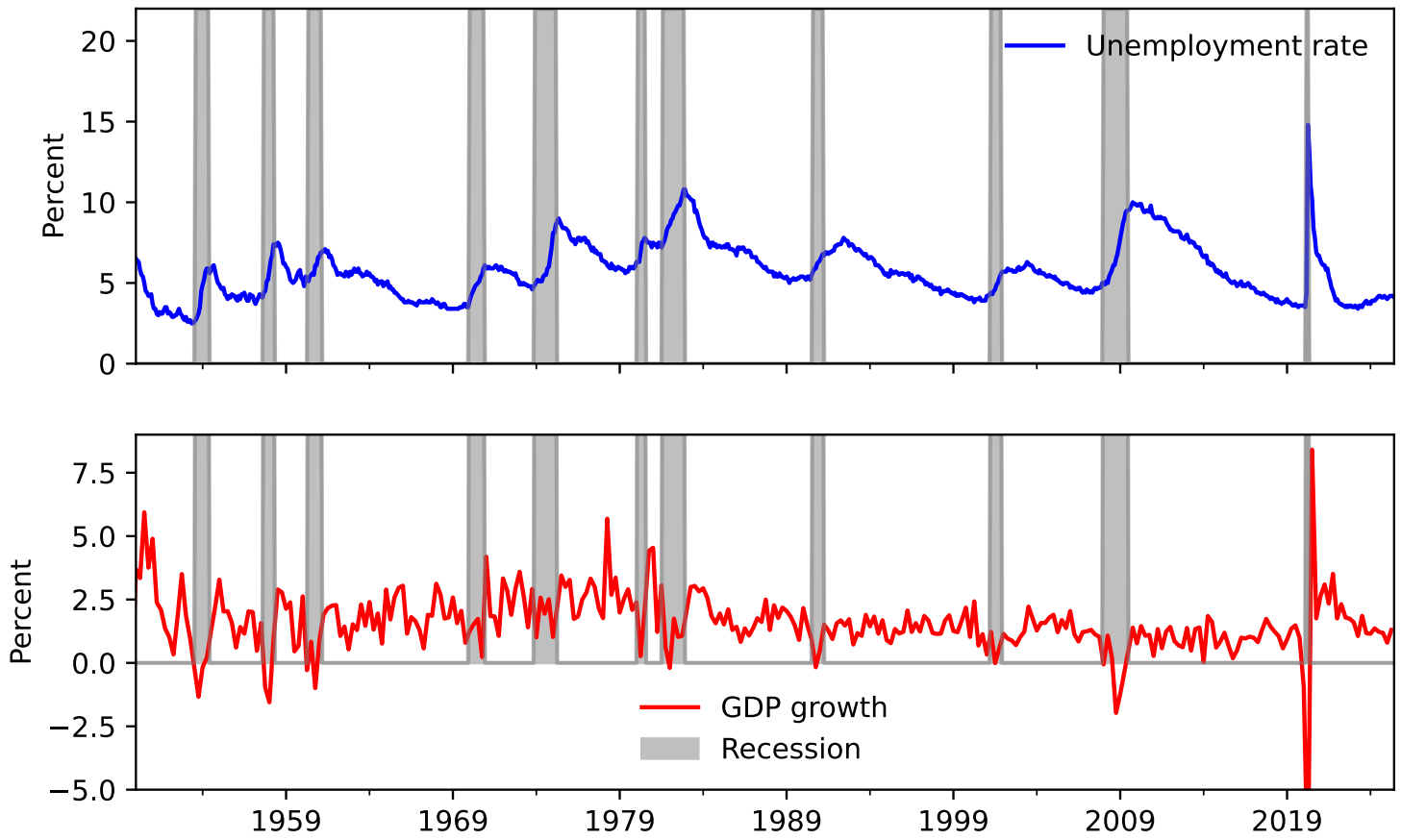
GDP across countries – OECD



GDP across countries – OECD

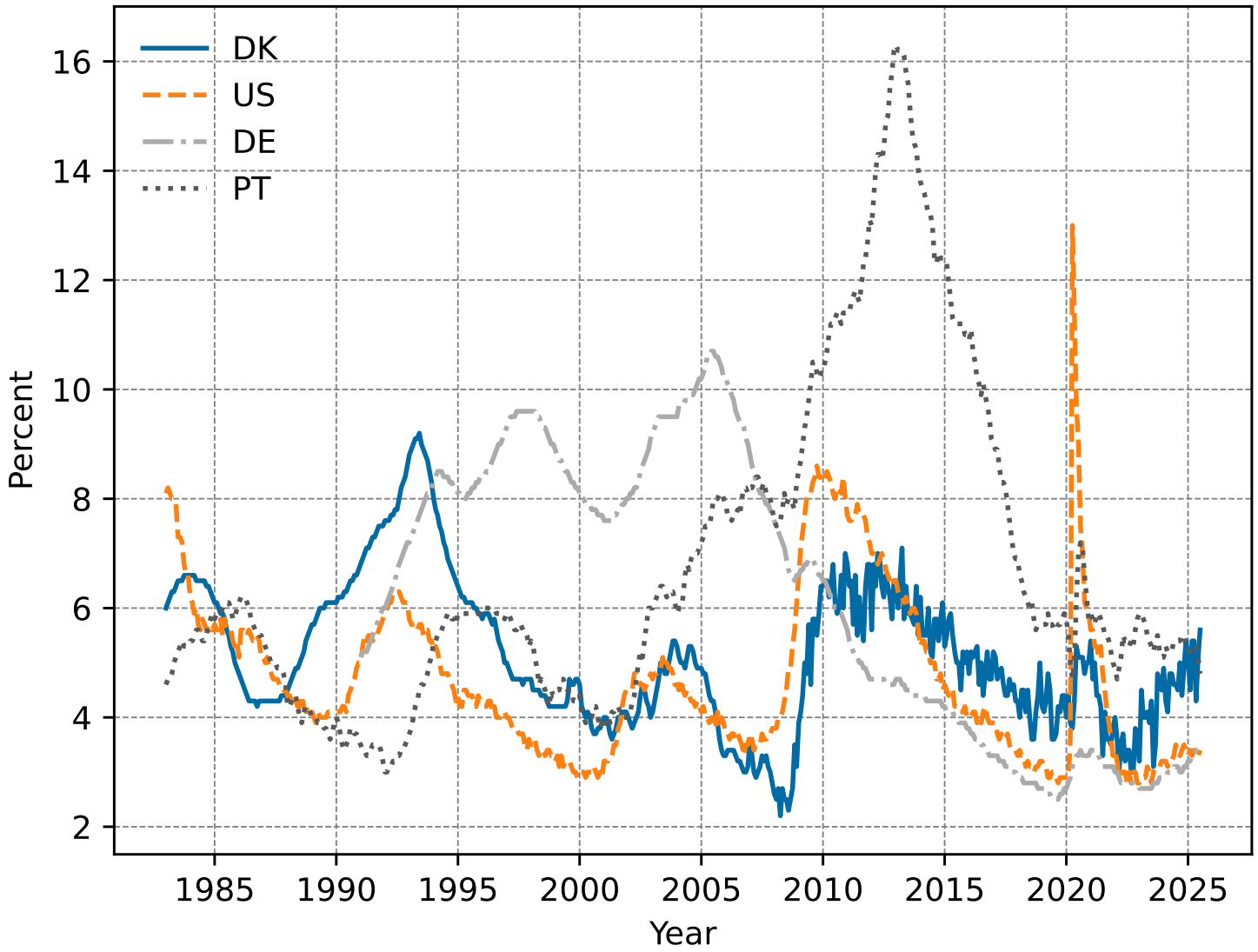


Unemployment and GDP growth in the US FRED

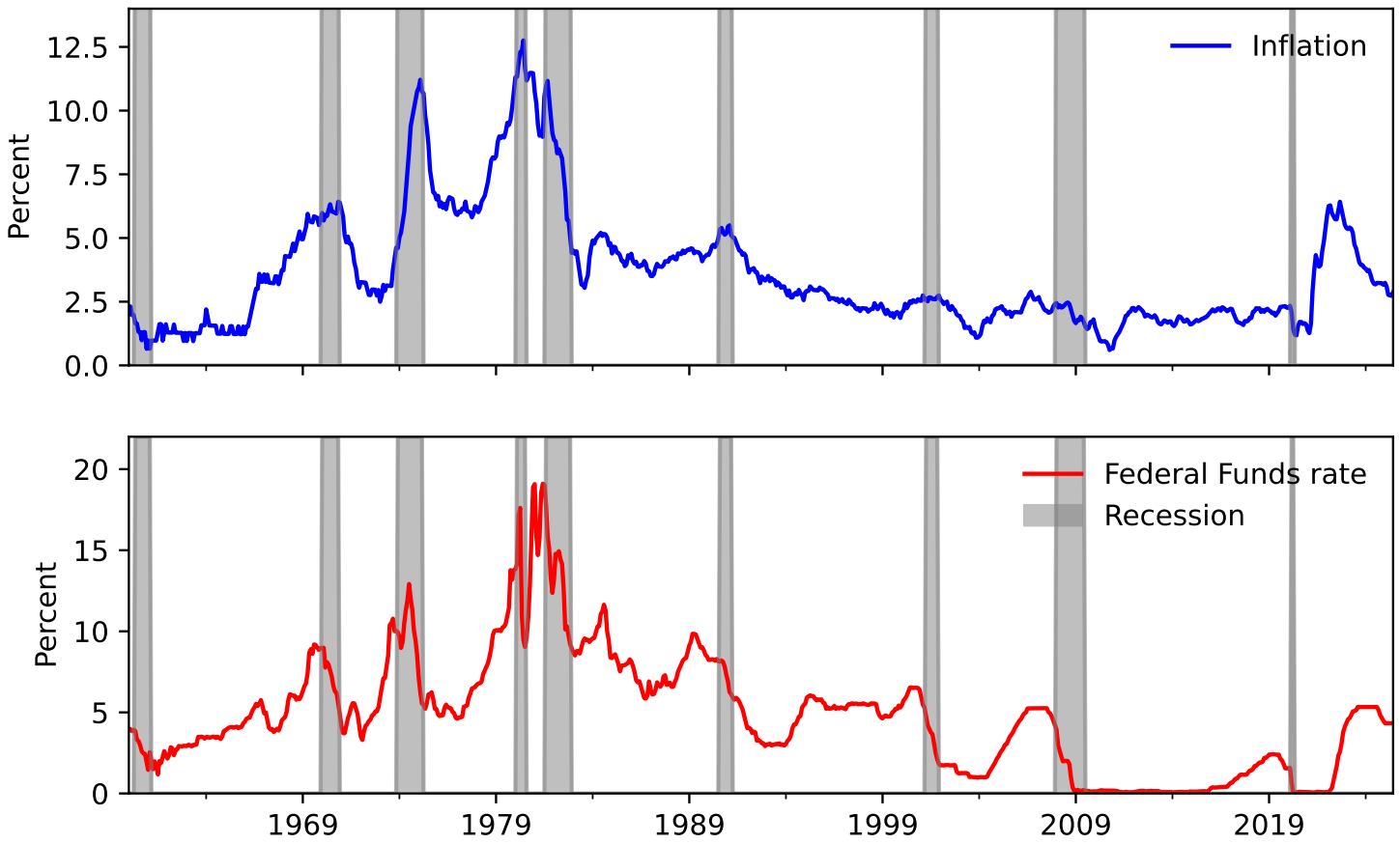


Unemployment in Europe

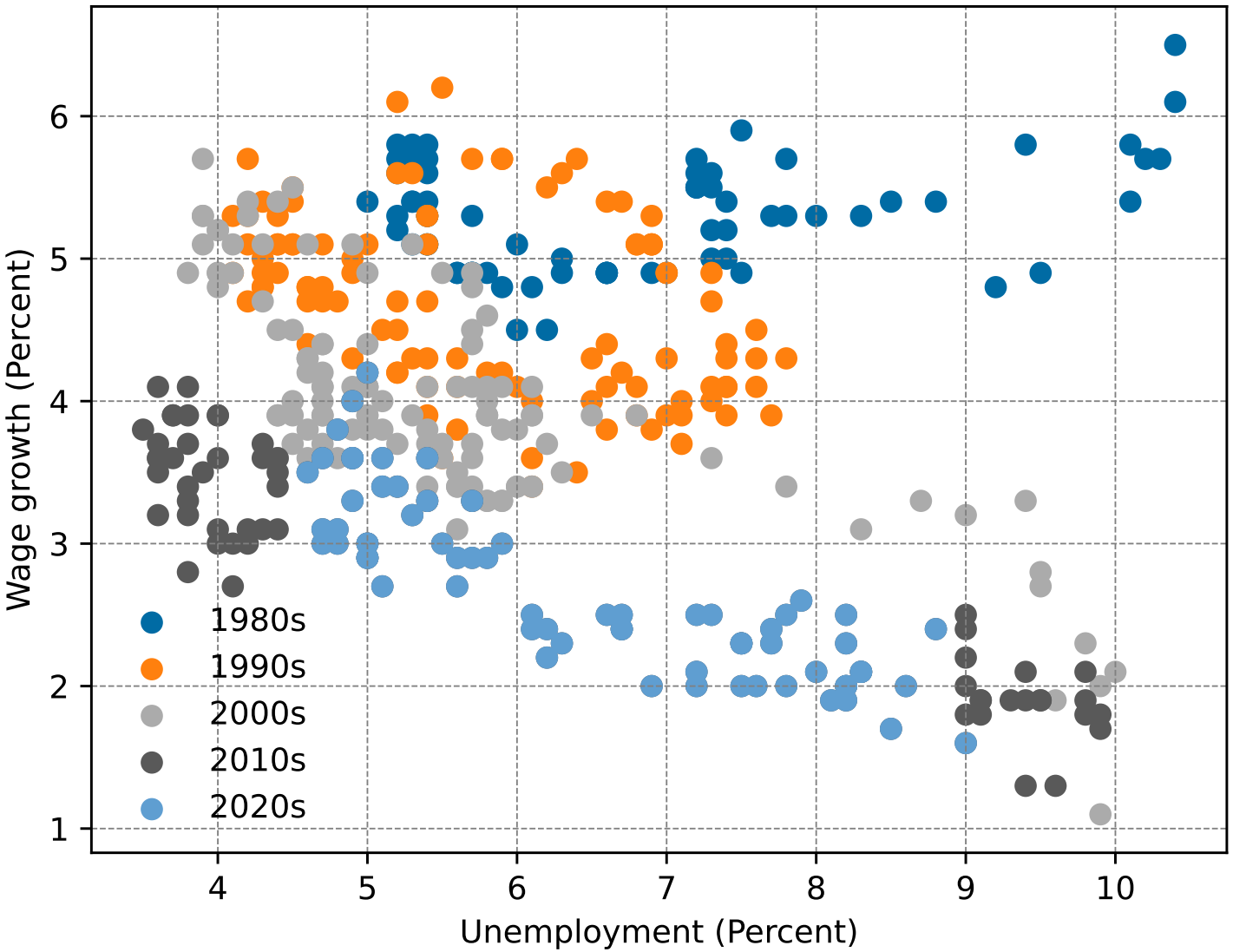
Eurostat



Inflation and interest rates in the US FRED

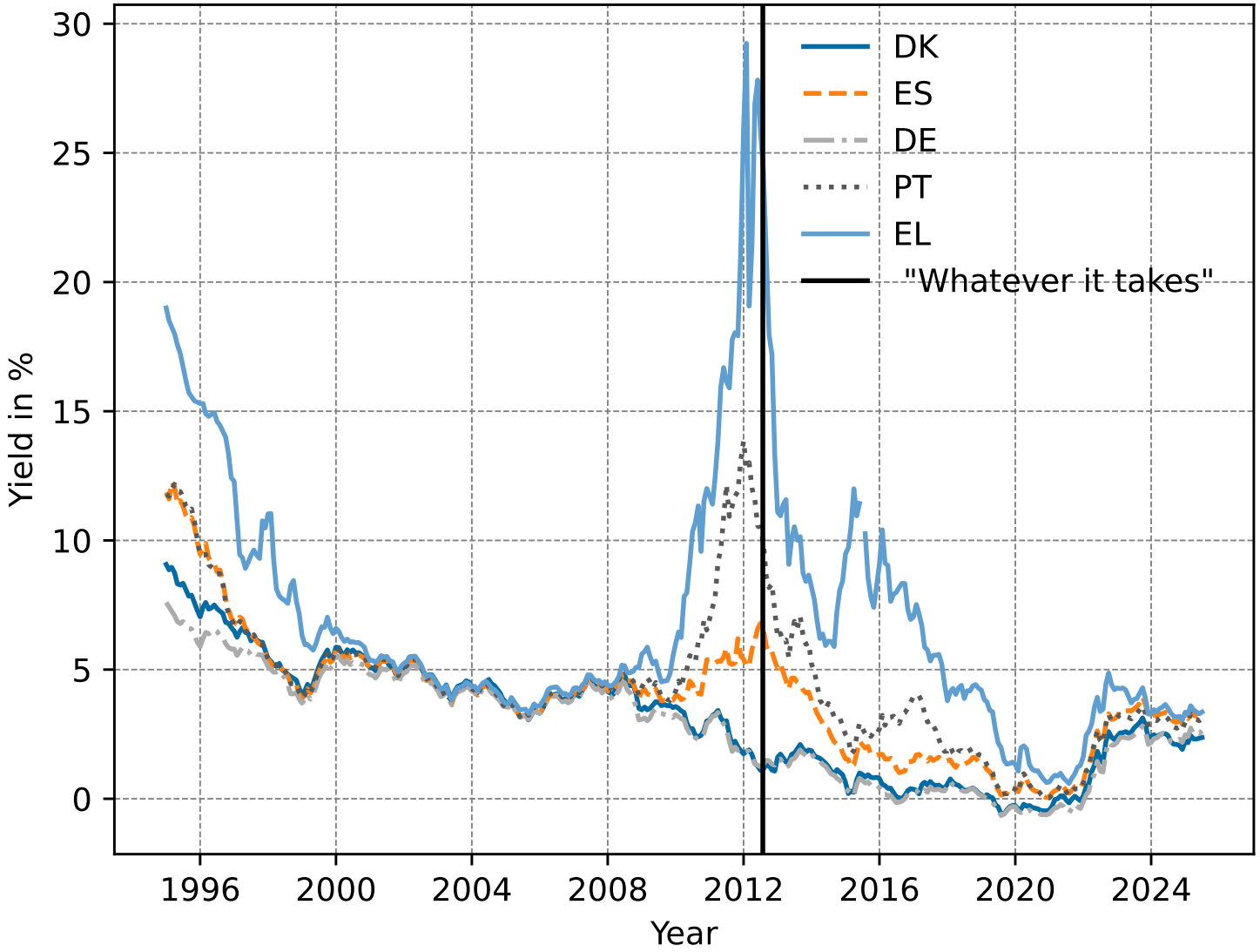


The US Phillips Curve FRED



The words of a European central banker

Eurostat



Macroeconomic questions

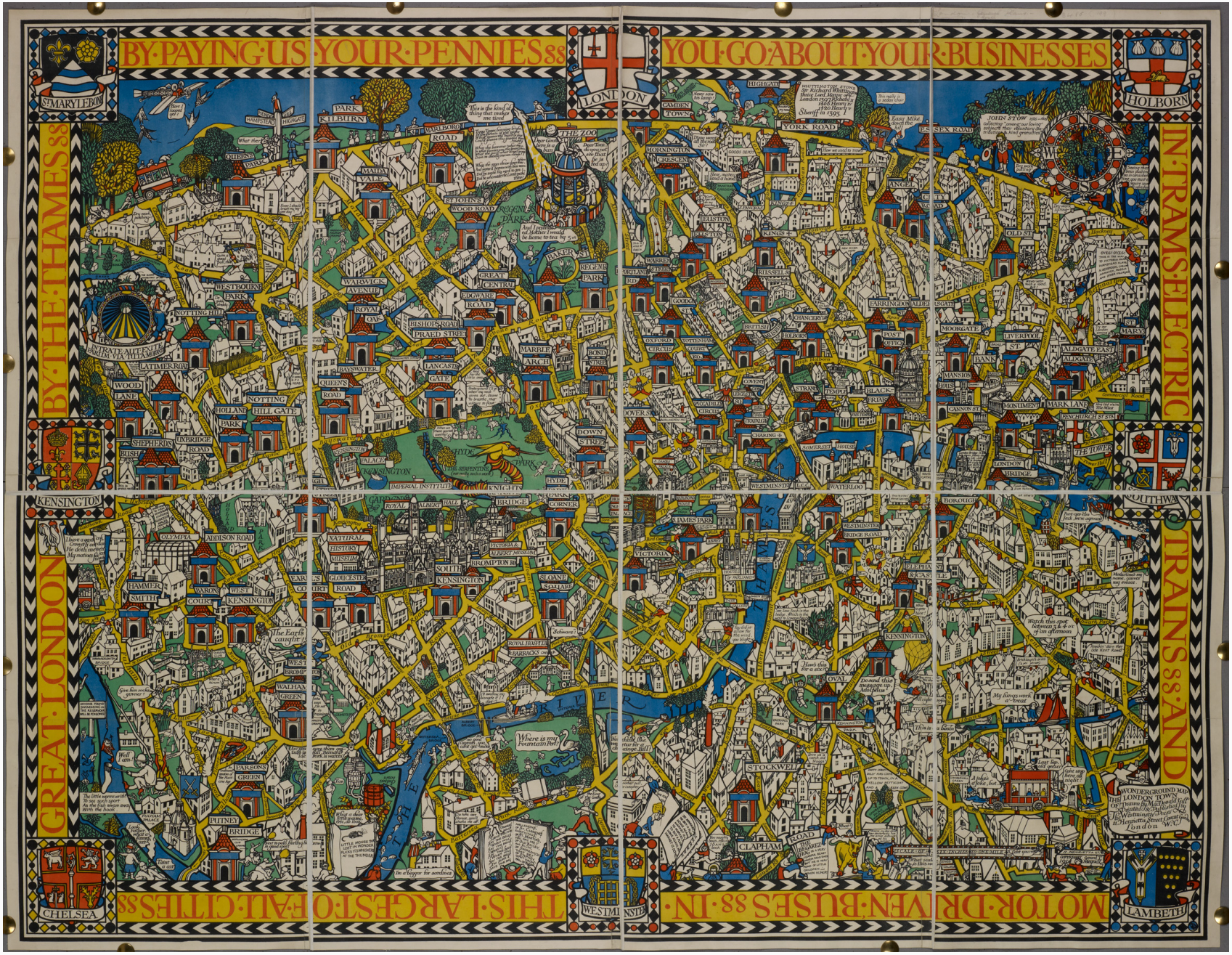
Very rare to get experiments at the aggregate level (in advance)

- How much more do people work if their wages rise?
- What if the government pressures the central bank, reducing its independence?
- How does inflation move after the ECB raises interest rates? (And why?)
- What happens when the US imposes 25% tariffs on its biggest trading partners?

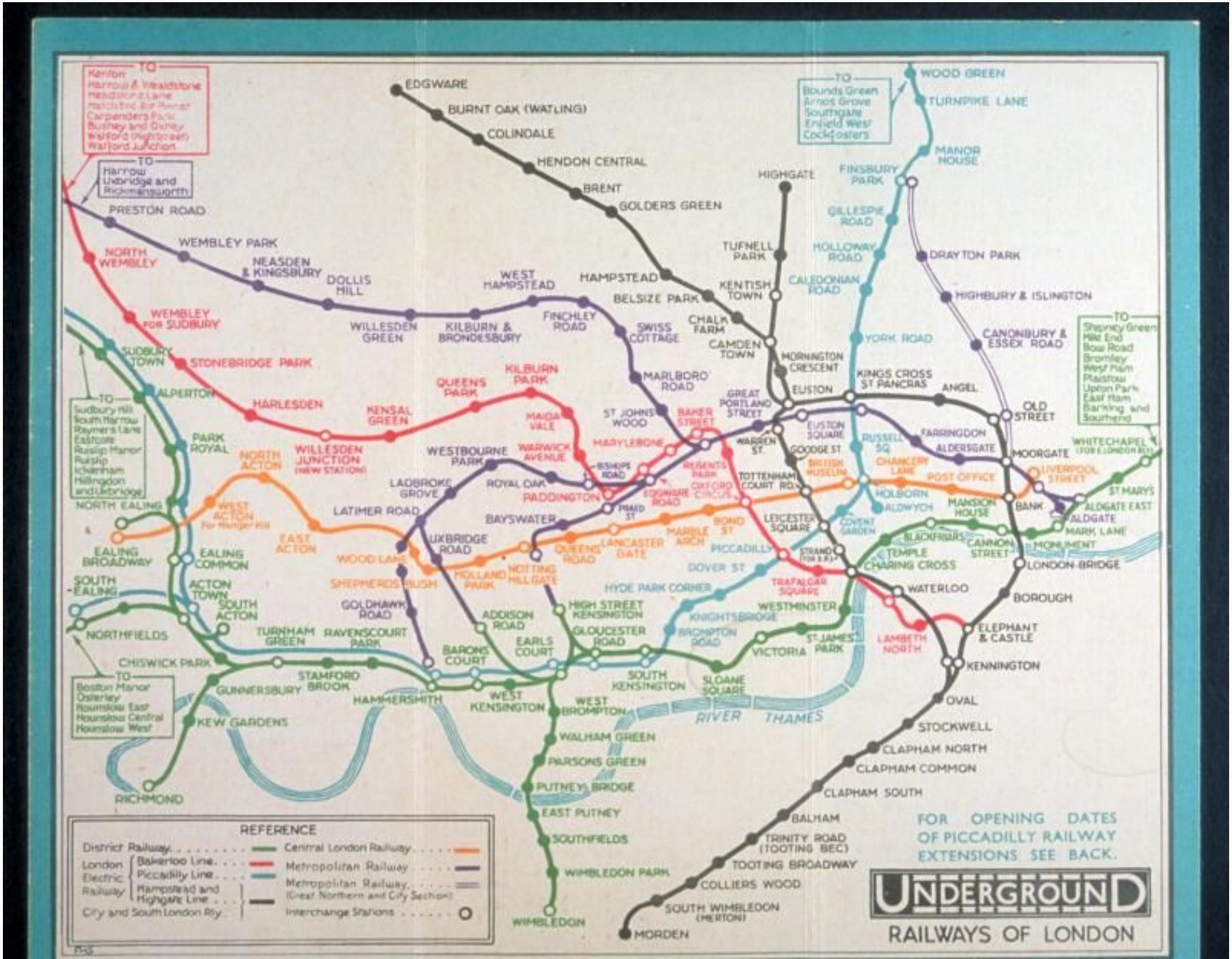
Empirics very helpful, but alone not enough

- Need frameworks to analyze the present and predict the future
- Especially if such an event has not happened before

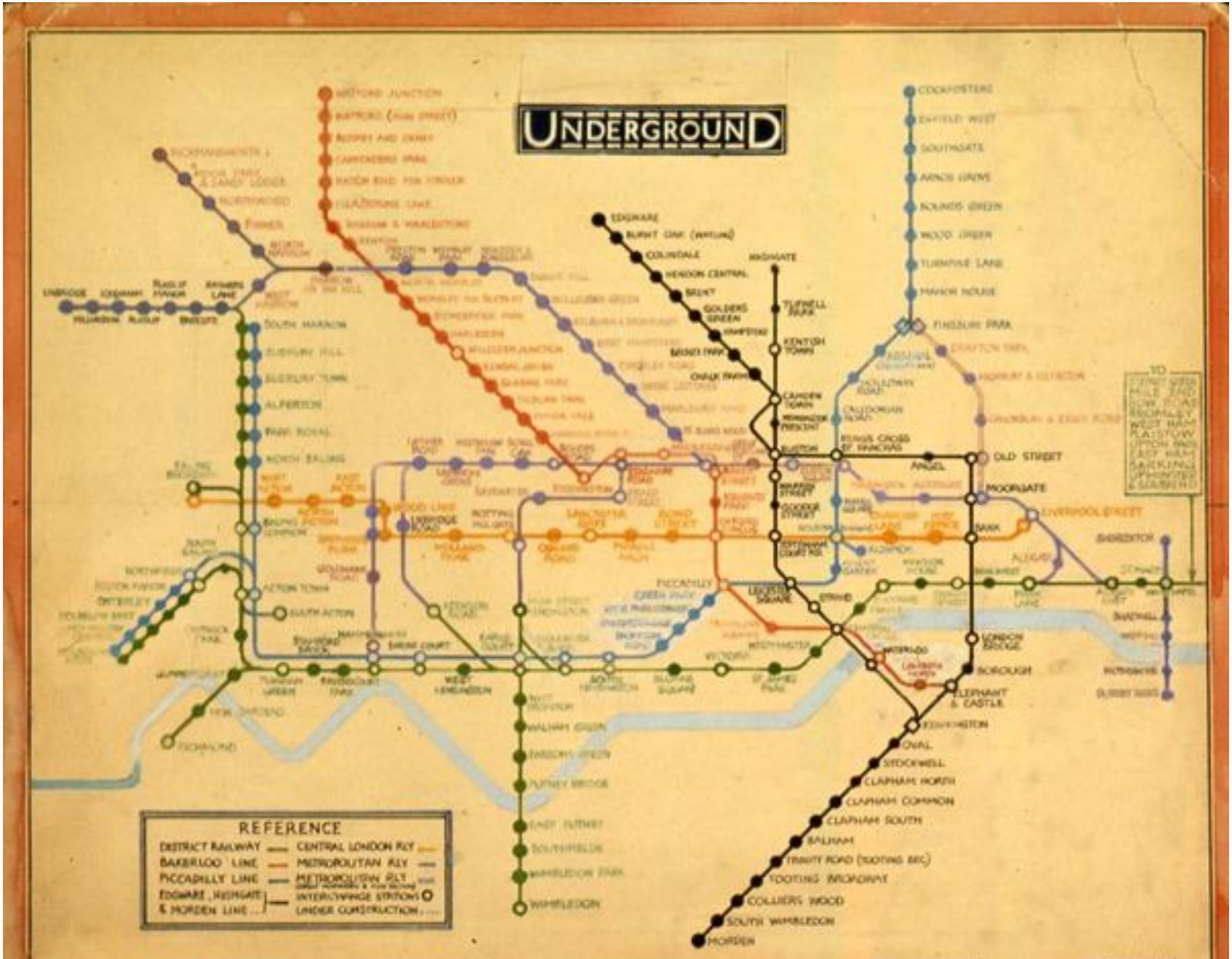
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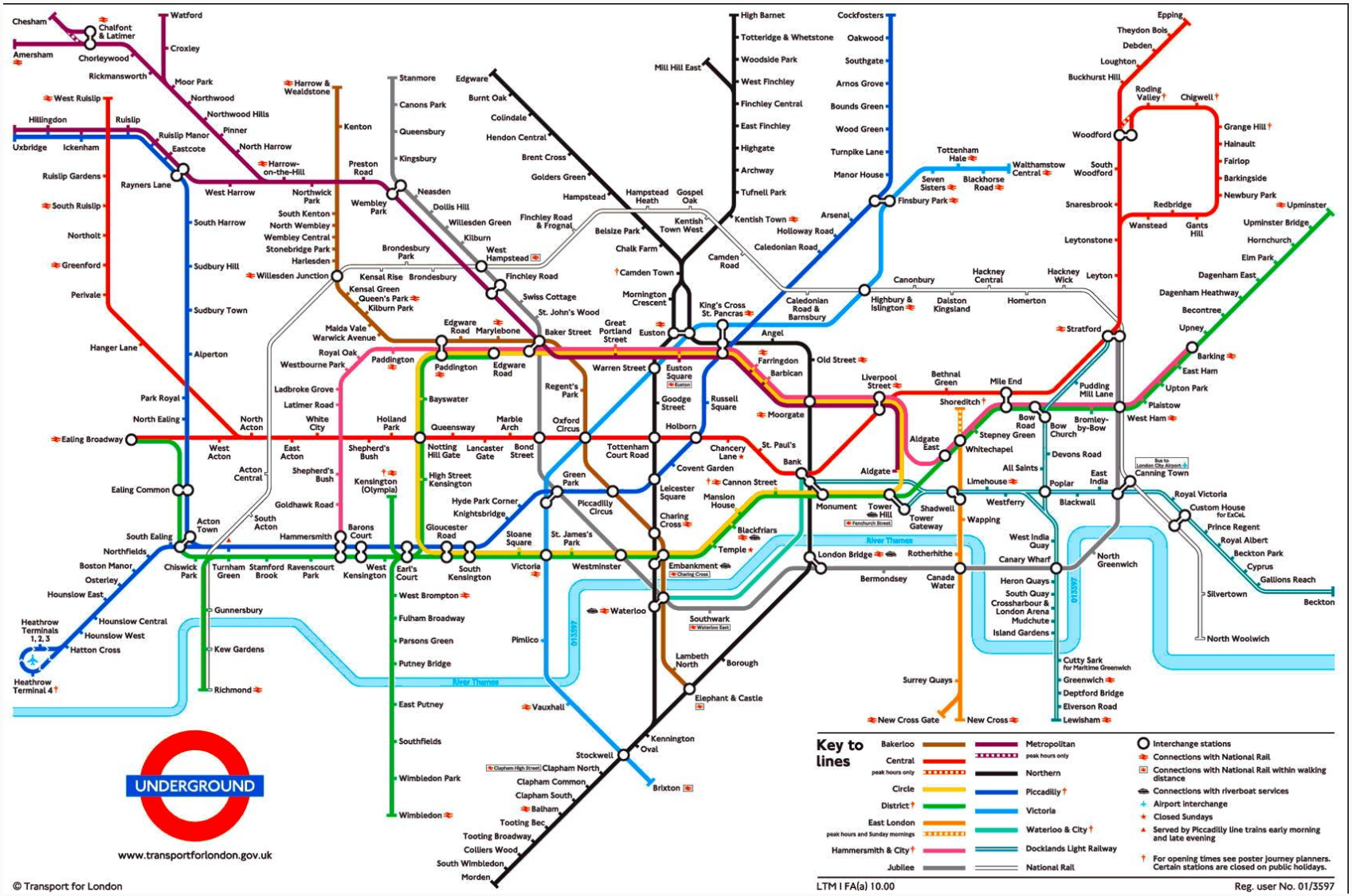
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UNDERGROUND
REFERENCE
DISTRICT RAILWAY
BAKERLOO LINE
PICCADILLY LINE
EDGWARE, HIGHGATE & MORDEN LINE
CENTRAL LONDON RLY
METROPOLITAN RLY
METROPOLITAN RLY
INTERCHANGE STATIONS
UNDER CONSTRUCTION
EDGWARE
BURNT OAK (WATLING)
COLINDALE
HENDON CENTRAL
BRENT
GOLDERS GREEN
HAMPSTEAD
HIGHGATE
TUFFNELL PARK
KENTISH TOWN
CAMDEN TOWN
EUSTON
KING'S CROSS ST. PANCRAS
OLD STREET
MOORGATE
BANK
LIVERPOOL STREET
BARBICAN
ANGEL
OXFORD CIRCUS
PICCADILLY CIRCUS
LEICESTER SQUARE
STRAND
CHARING CROSS
EMBANKMENT
WATERLOO
LONDON BRIDGE
BOROUGH
ELEPHANT & CASTLE
KENNINGTON
OVAL
STOCKWELL
CLAPHAM NORTH
CLAPHAM COMMON
CLAPHAM SOUTH
BALHAM
TRINITY ROAD (TOOTING BEC)
TOOTING BROADWAY
COLLIERS WOOD
SOUTH WIMBLEDON
MORDEN
WIMBLEDON
WIMBLEDON PARK
SOUTHFIELDS
EAST PUTNEY
PUTNEY BRIDGE
PARSONS GREEN
WALHAM GREEN
FULHAM BROADWAY
WEST BROMPTON
SLOANE SQUARE
SOUTH KENSINGTON
GLOUCESTER ROAD
KENSINGTON (OLYMPIA)
EARL'S COURT
WEST KENSINGTON
BARONS COURT
HAMMERSMITH
TURNHAM GREEN
STAMFORD BROOK
RAVENSCOURT PARK
SHEPHERD'S BUSH
GOLDHAWK ROAD
SOUTH ACTON
ACTON TOWN
NORTH ACTON
EALING BROADWAY
WEST ACTON
EAST ACTON
NORTH EALING
PARK ROYAL
ALPERTON
SUDBURY TOWN
SUDBURY HILL
SOUTH HARROW
RAYNERS LANE
RUISLIP MANOR
RUISLIP
EASTCOTE
PINNER
NORTHWOOD
NORTHWOOD HILLS
MOOR PARK
CROXLEY
WATFORD JUNCTION
WATFORD (HIGH STREET)
CHESHAM
AMERSHAM
CHALFONT & LATIMER
CHORLEYWOOD
RICKMANSWORTH
MOOR PARK
NORTHWOOD
PINNER
NORTH HARROW
HARROW & WEALDSTONE
KENTON
QUEENSBURY
KINGSBURY
WEMBLEY CENTRAL
NORTH WEMBLEY
SOUTH KENTON
NORTH WEMBLEY
WEMBLEY PARK
SUDBURY HILL HARROW
SUDBURY TOWN
ALPERTON
PARK ROYAL
NORTH EALING
WILLESDEN JUNCTION
KENSAL GREEN
QUEEN'S PARK
KILBURN PARK
MAIDA VALE
WARWICK AVENUE
CLIFTON HILL
ST. JOHN'S WOOD
BAKER STREET
REGENT'S PARK
OXFORD CIRCUS
PICCADILLY CIRCUS
GREEN PARK
HYDE PARK CORNER
KNIGHTSBRIDGE
SOUTH KENSINGTON
SLOANE SQUARE
VICTORIA
WESTMINSTER
EMBANKMENT
CHARING CROSS
LEICESTER SQUARE
COVENT GARDEN
HOLBORN
TOTTENHAM COURT ROAD
GOODGE STREET
WARREN STREET
EUSTON
CAMDEN TOWN
MORNINGTON CRESCENT
CHALK FARM
ROYAL OAK
PADDINGTON
EDGWARE ROAD
MARBLE ARCH
BOND STREET
OXFORD CIRCUS
REGENT'S PARK
BAKER STREET
MARYLEBONE
LANCASTER GATE
QUEENSWAY
NOTTING HILL GATE
BAYSWATER
WOOD LANE
LATIMER ROAD
WHITE CITY
HOLLAND PARK
SHEPHERD'S BUSH
LADBROKE GROVE
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NORTH ACTON
EAST ACTON
WE

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Understanding/explaining macroeconomic phenomena formally

Models

- Simplified description of the world
- Useful analytical tools

Advantages

- All assumptions clearly stated
- Rigorous representation of the world
- Focus on the most important aspects
- Counterfactuals can easily be understood

Disadvantages

- Simplifications

Main departures from previous courses

Mathematical formalism

- Explain macroeconomic phenomena using common language
- Learn new analytical tools
- Individual constrained optimization

Microfoundations

- Models are built from the ground up
- Consistent at micro- and macro-level
- Can be compared to micro-data

$\Longrightarrow$ we get the answers right for the right reasons

Main models to be covered

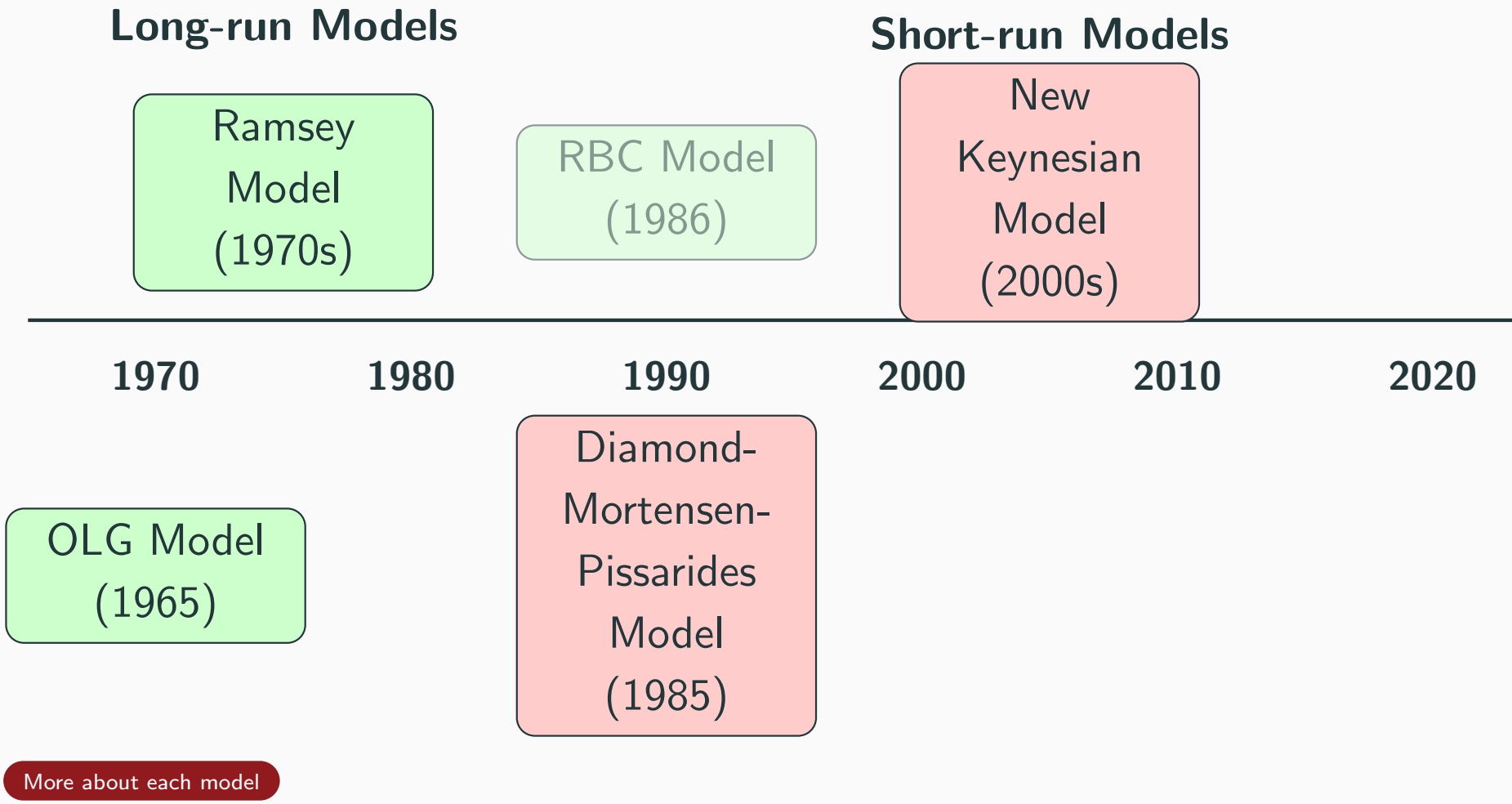
Long-run models

- Ramsey model (aka the neoclassical growth model)
- Overlapping generations model (OLG)

Business cycles

- The real business cycle (RBC) model
  - Open Economy Macro (at the end)
- Diamond-Mortensen-Pissarides (DMP) model of the labor market
- Nominal rigidities
- The New Keynesian (NK) model of central banking

Main models to be covered



More about each model

Brief history of macroeconomic thought

Keynes' General Theory of employment, interest & money

Beginning of modern macroeconomic theory

- Response to Great Depression
- Attempting to formulate policy responses for the future

Major innovation

- Business cycles are demand driven
- Analytical reasoning, but without formalizing in mathematical terms
- Makes real vs nominal interaction central for the first time
- Markets may not clear automatically
- Involuntary unemployment

Neoclassical-Keynesian synthesis

Formalization of Keynes' ideas

- Hicks (Nobel 1972): IS-LM model
- Samuelson (Nobel 1970): established MIT as a leading center of economics

Major innovation

- Keynes is relevant for the short-run, classical theory is a long-run concept
- Turning Keynes' writings into policy prescriptions
- IS-LM model
- Inclusion of the labor market into the model ($\Longrightarrow$ Phillips curve)

(Actually more of a coexistence than a synthesis)

Monetarism

Milton Friedman (University of Chicago)

- Nobel Laureate of 1976

Major innovation

- "Inflation is always and everywhere a monetary phenomenon" (Friedman & Schwartz, 1963)
- Monetary policy can have unintended consequences
- Great Depression was worsened by lack of money supply
- Permanent income hypothesis
- Inflation expectations adjust endogenously to policy

The new classical counter-revolution

Fresh water economics

- Lucas (Nobel 1995), Sargent (Nobel 2011), Wallace
- Kydland, Prescott (both Nobel 2004)

Major innovation

- Flexible prices, even in the short-run + Walrasian market clearing
- Rational expectations under uncertainty
- Policy ineffectiveness proposition
- Microfoundations, stochastic modeling of agents, optimization
- Business cycles are mainly supply-driven

The New Keynesian reconstruction

Salt water economics

- Blanchard, Akerlof, Yellen, Rotemberg, Kiyotaki
- Woodford, Gali

Major innovation

- Acceptance of rational expectations and microfoundations
- Monopolistic competition
- Nominal rigidities (i.e., sticky prices)
- Short-run fluctuations caused by supply and demand

Modern macroeconomics

Heterogeneity

- Increased computing power allows for more heterogeneity
- Modeling inequality across households

Data

- Computing power allows for bigger datasets to test more complicated models

Imperfect-information rational expectations models

- Costly information acquisition, behavioral biases, etc

Climate economics

- Synthesis with environmental economics

Microfoundations

Microfoundations

Macroeconomic phenomena have to make microeconomic sense

- No more ad hoc assumptions about macroeconomic relationships
- Optimizing agents make *rational* decisions
- Utility functions and discounting of future consumption
- Aggregate up individual choices

Intertemporal consumption & savings choices (simple)

Assumptions

- Time is discrete (i.e., not continuous)
- There is a representative agent
- Two period model $t \in \{0, 1\}$
- Agents discount the future at factor β
- The agent can save in bonds (a_1) which pay interest rate $R(= 1 + r)$
- Each period, agents receive an endowment y_t, **which is known**
- The utility function is given by $u(c_t)$

No general equilibrium yet

- Assume R is given exogenously ("Partial equilibrium")

Solving a two-period consumption/savings problem

Maximization problem

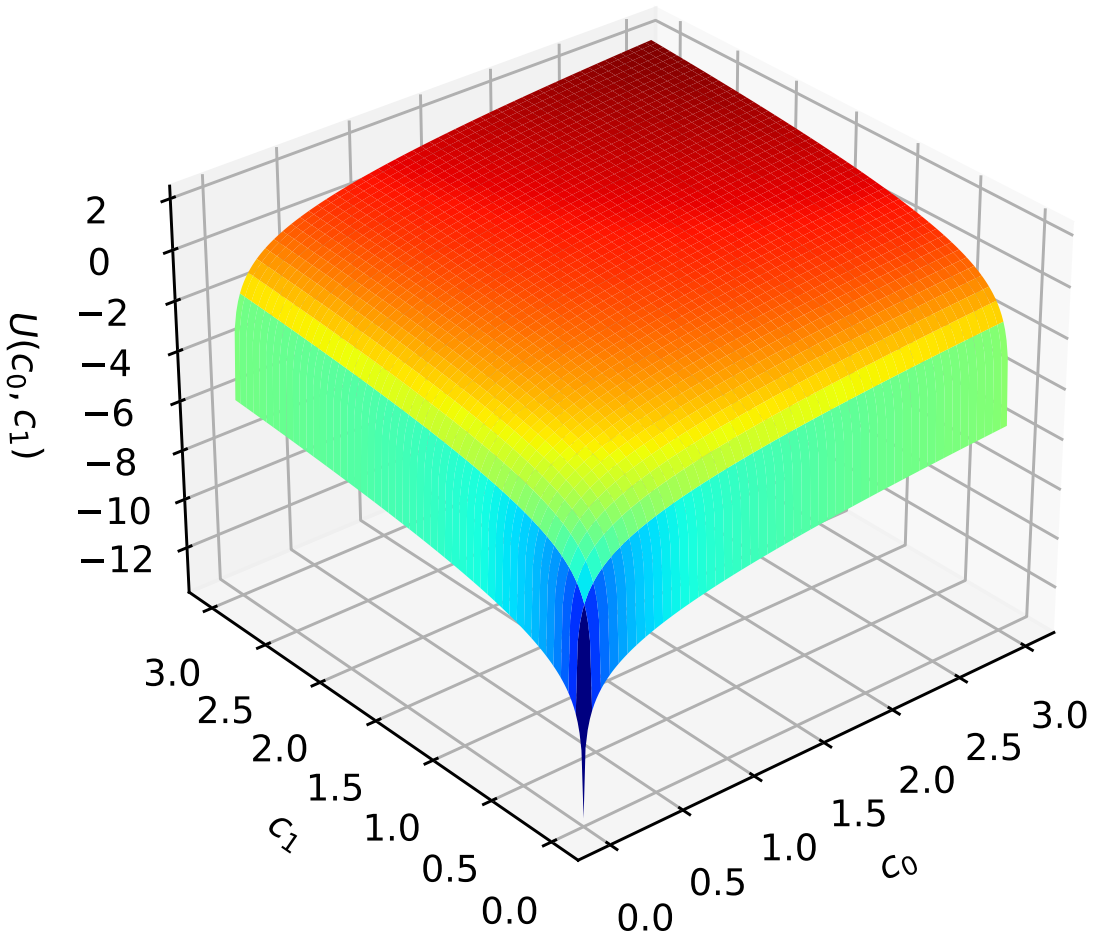
$$\begin{aligned} \max_{c_0, c_1} \quad & U = u(c_0) + \beta u(c_1) \\ \text{subject to} \quad & c_0 + a_1 = y_0 \\ & c_1 = y_1 + (1 + r)a_1 \end{aligned}$$

- Budget constraints are "dynamic" $\Longrightarrow$ influence each other
- Solution: combine and substitute

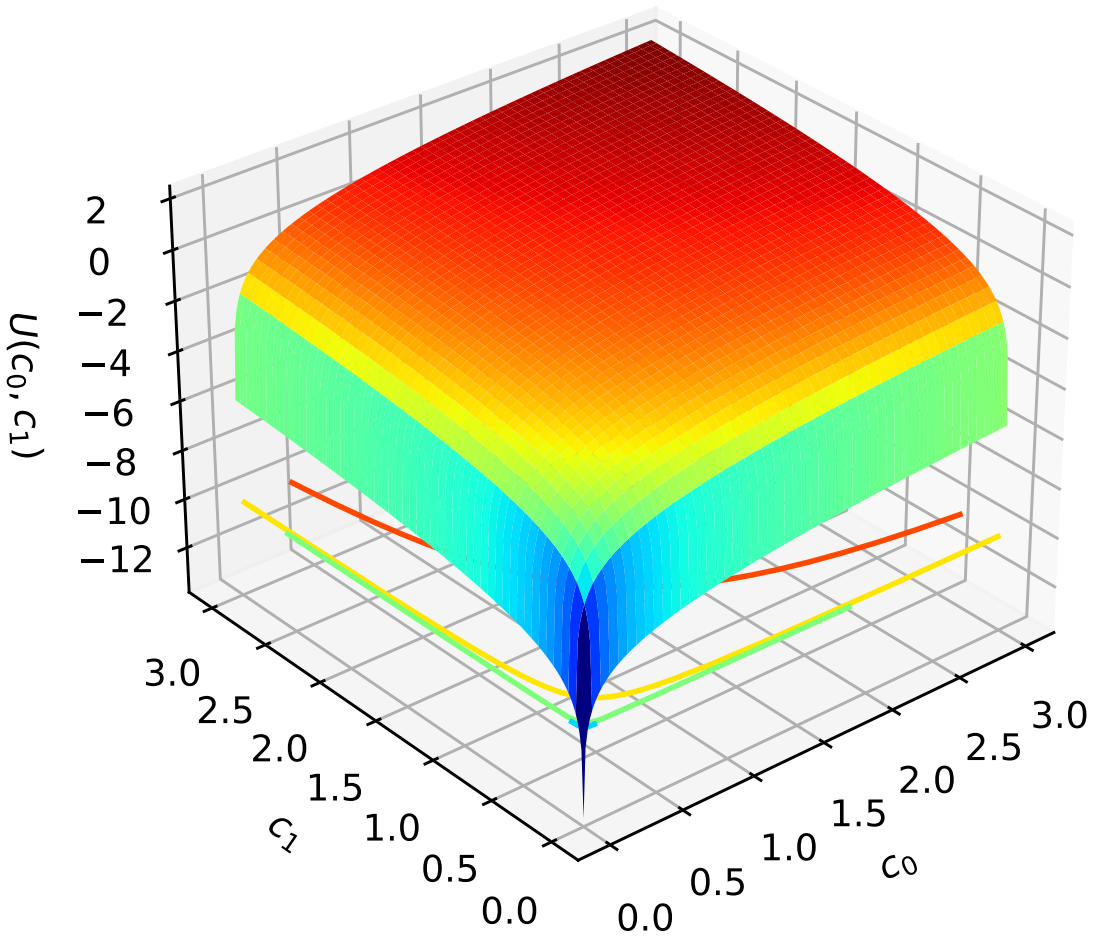
$$c_0 = y_0 - \frac{c_1 - y_1}{1 + r}$$

$\rightarrow$ will use this to substitute into utility function

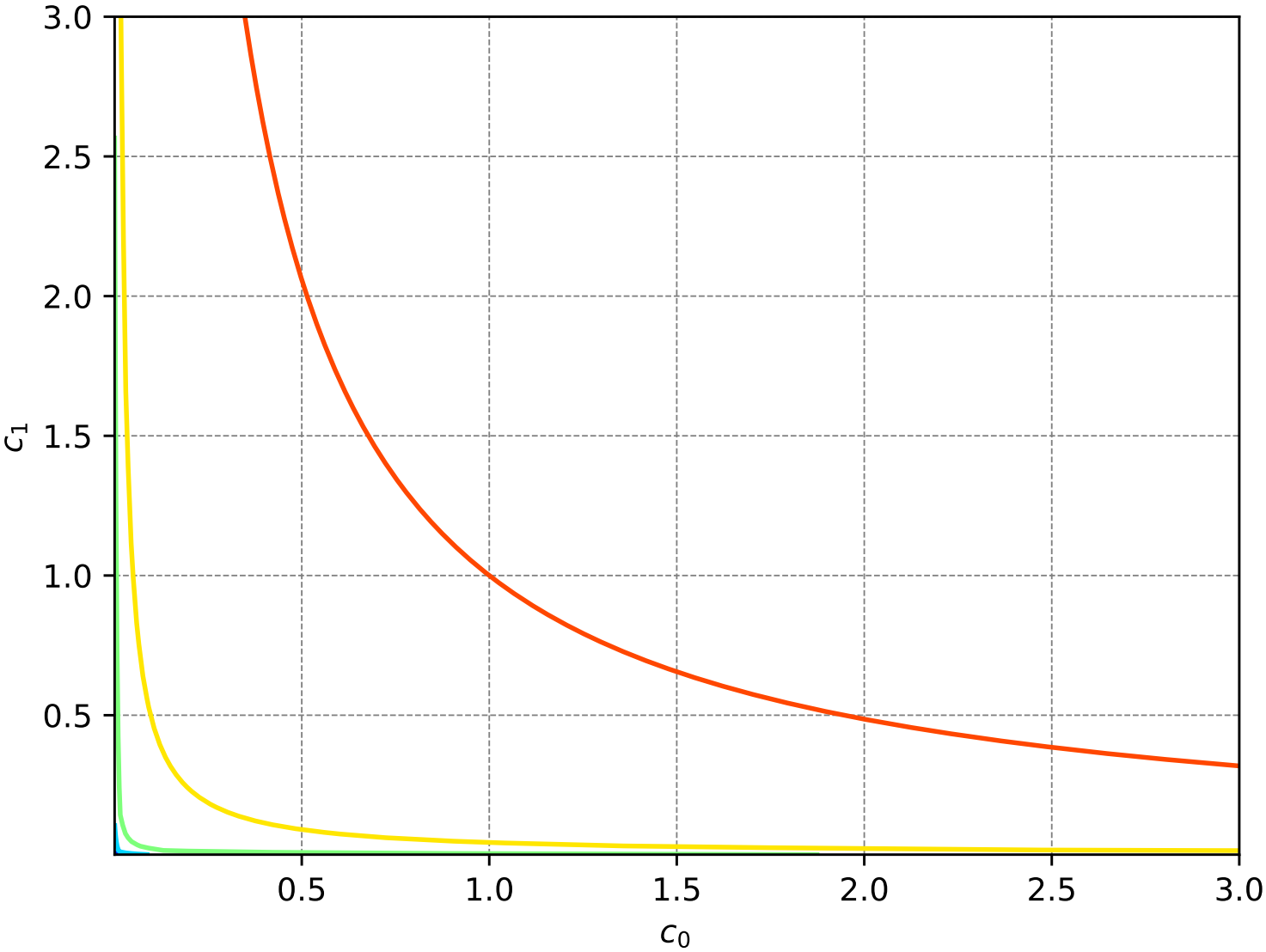
Preferences – graphical



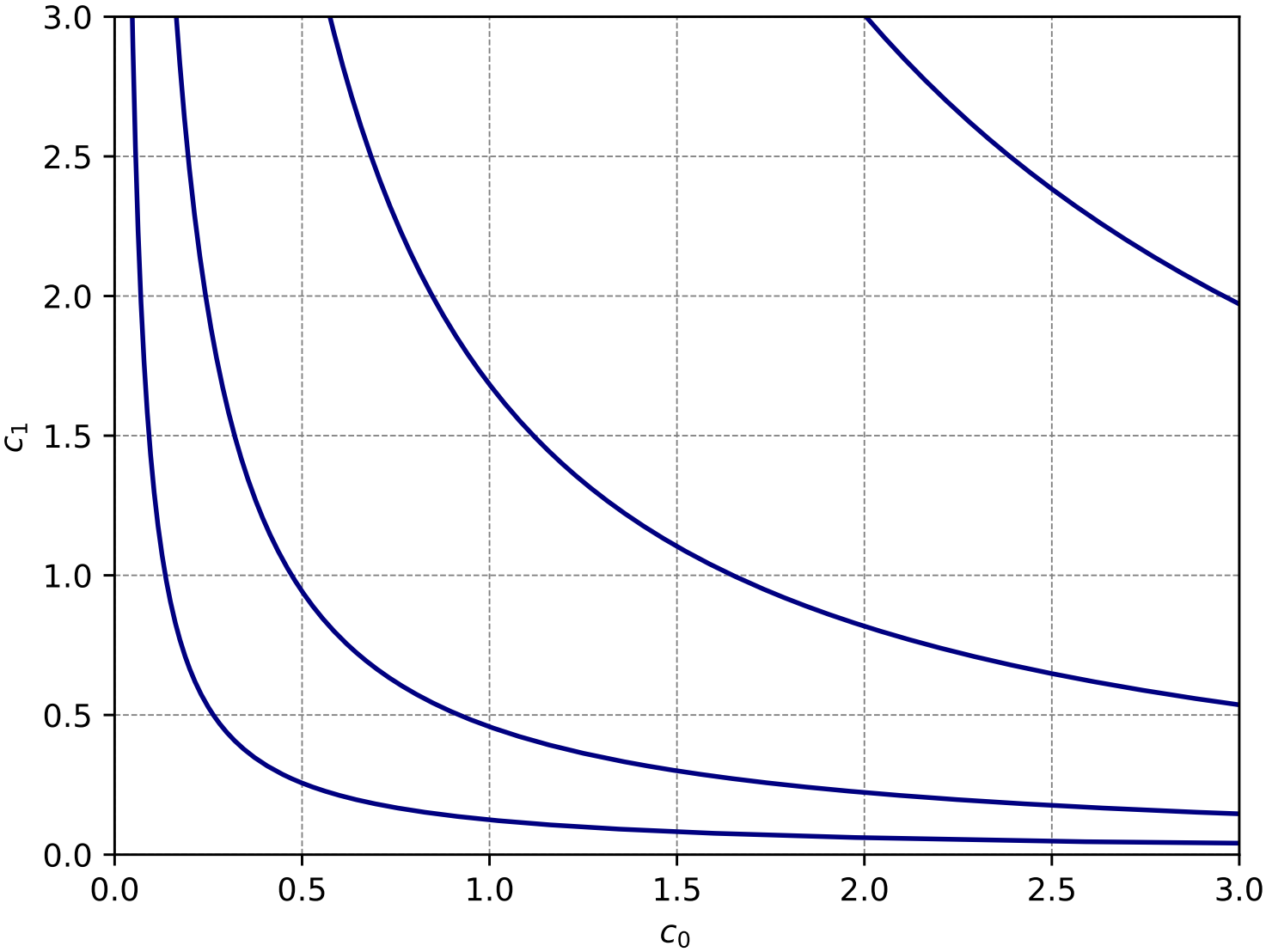
Preferences – graphical



Preferences – graphical



Preferences – graphical



Constrained optimization

Maximization problem

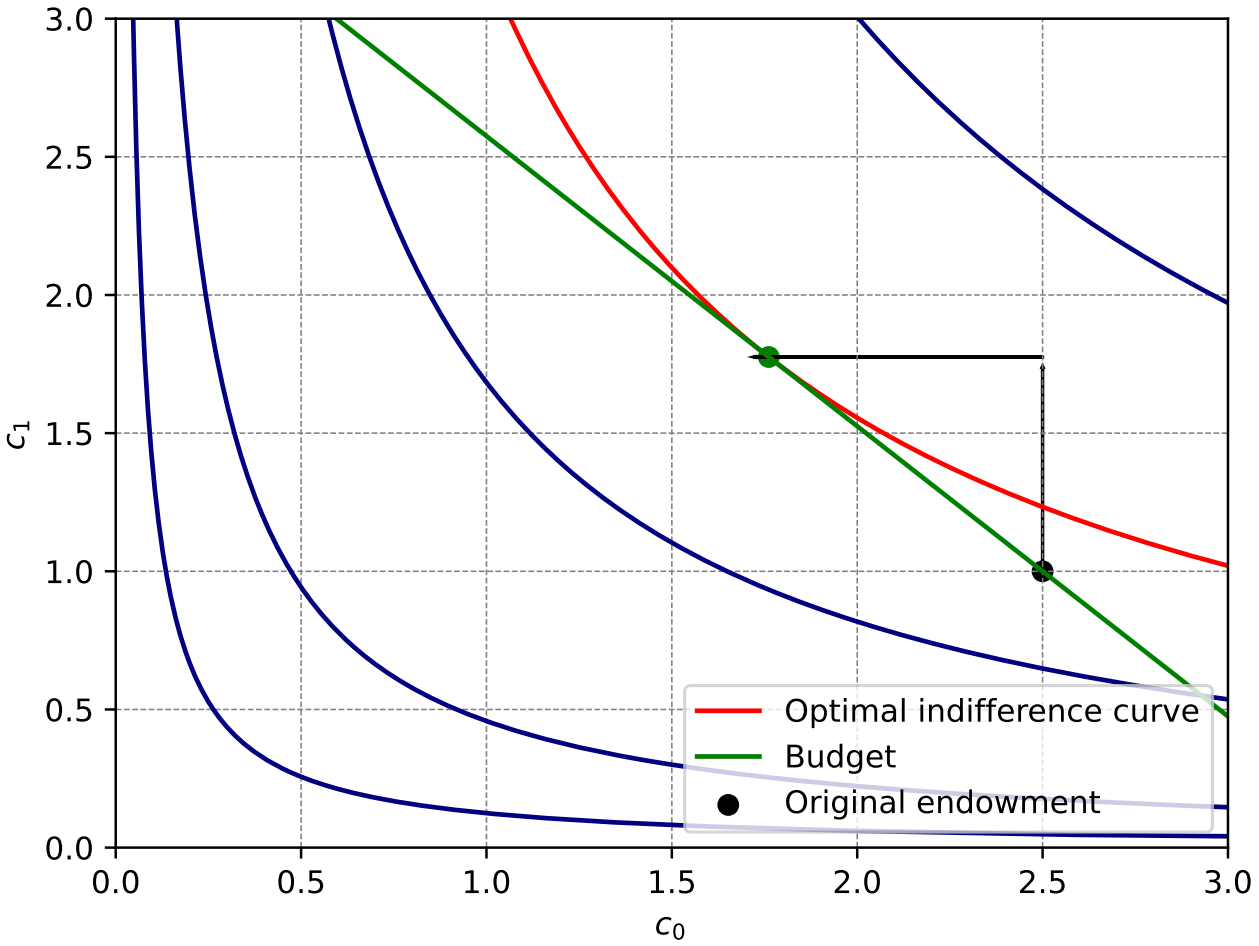
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First order condition for optimality

$$\beta u'(c_1) = \frac{1}{1+r} u' \underbrace{\left(y_0 - \frac{c_1 - y_1}{1+r} \right)}_{c_0}$$

$$\underbrace{u'(c_0) = \beta(1+r)u'(c_1)}_{\text{Euler equation}} \text{ and } \underbrace{c_1 = y_1 + (1+r)(y_0 - c_0)}_{\text{Budget constraint}}$$

Solution – Graphical (decreasing endowment path)



- Agents save to smooth consumption

Intuition

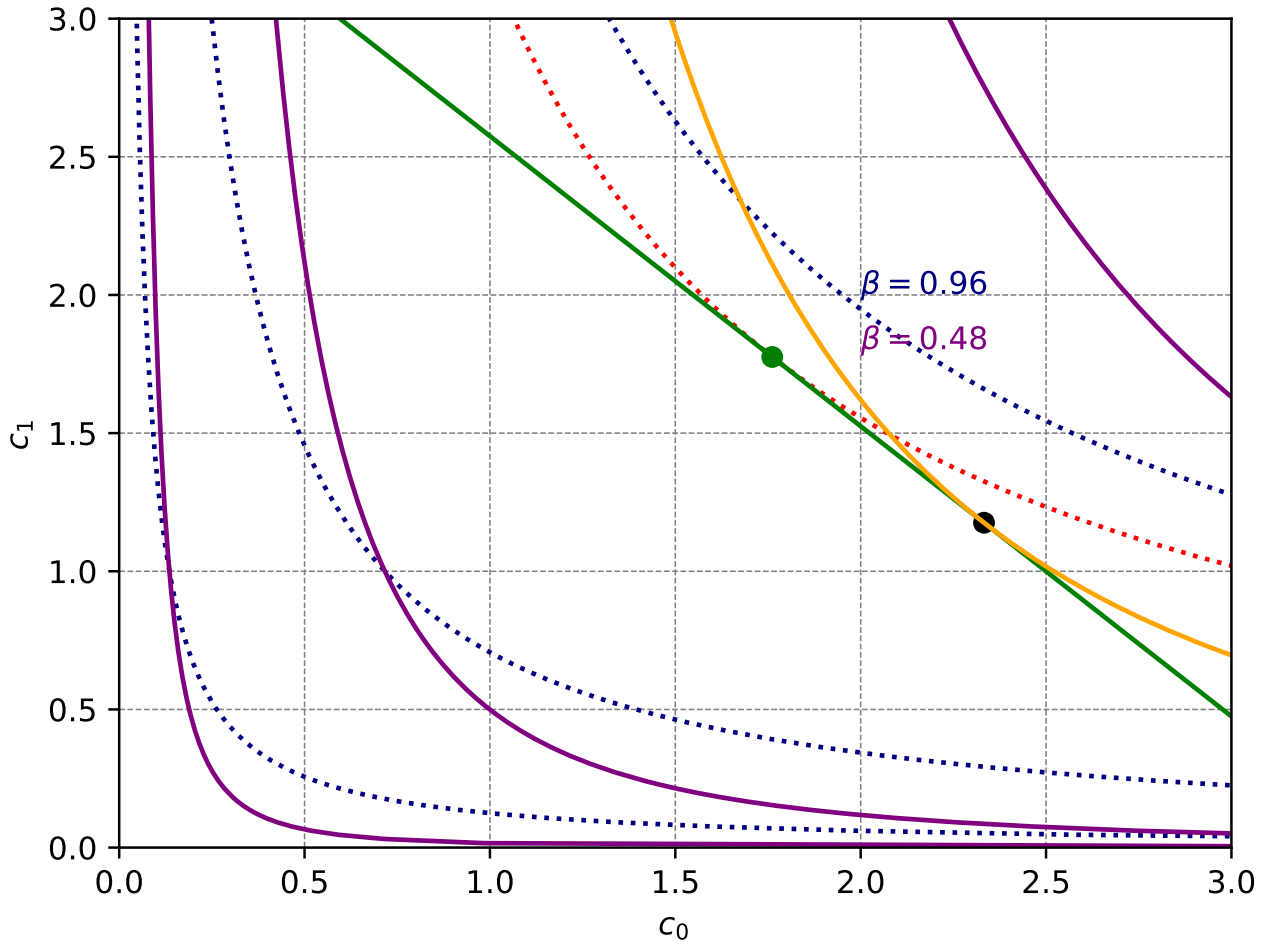
Euler equation

$$u'(c_0) = \beta(1+r)u'(c_1) \iff \frac{u'(c_0)}{\beta u'(c_1)} = (1+r)$$

Tradeoff between marginal utility today and tomorrow is governed by

- Endowment path y_t
- Patience β
- Shape of the utility function
- Interest rate r

Patience vs impatience – $u'(c_0) = \beta(1+r)u'(c_1)$



- Impatient consumers consume more today ($\beta \downarrow \rightarrow c_0 \uparrow, \quad c_1 \downarrow$)

Utility function

Crucial assumptions

- Continuously differentiable and displays positive but decreasing marginal utility: $\lim_{c\to\infty} u'(c) = 0$
- Consumption is essential: $\lim_{c\to 0} u'(c) = \infty$

The workhorse – Constant Relative Risk Aversion (CRRA)

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}; \quad \sigma > 0; \sigma \neq 1$$

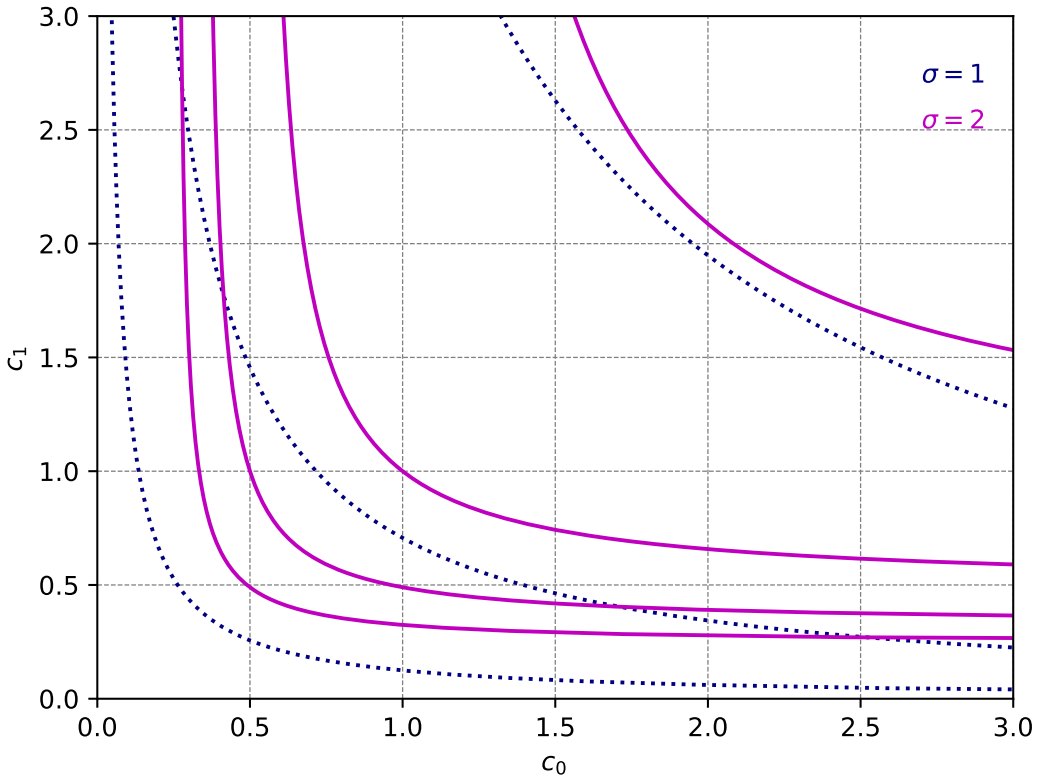
- σ is called the coefficient of relative risk aversion
- $1/\sigma$ is the intertemporal elasticity of substitution between two periods ($\sigma \to \infty \implies$ Leontief across periods)
- As $\sigma \to 1$, one can show that $u(c) = \ln(c)$

CRRA utility

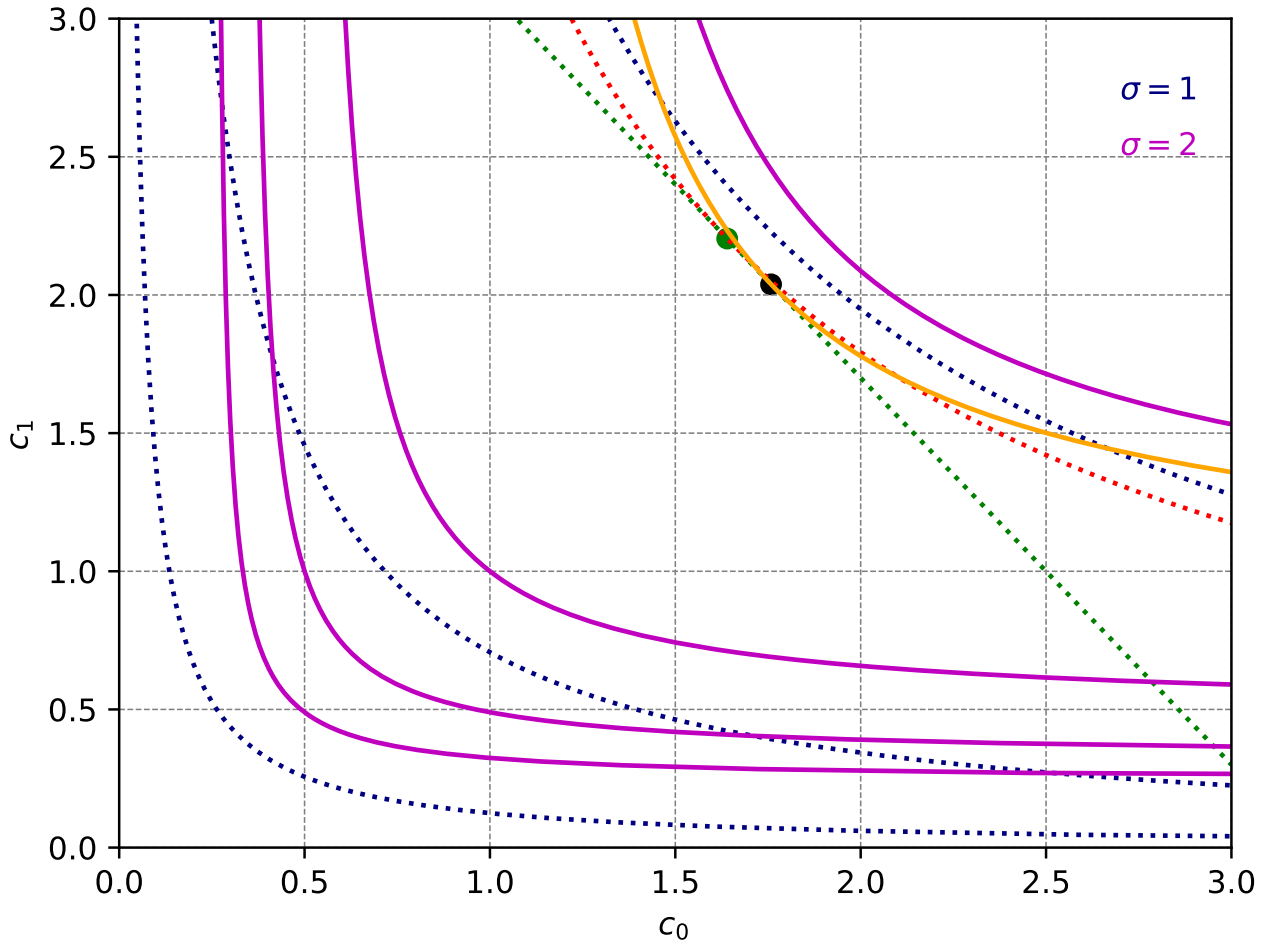
Euler equation

$$c_0^{-\sigma} = \beta(1+r)c_1^{-\sigma} \iff \left(\frac{c_0}{c_1}\right)^{-\sigma} = \beta(1+r)$$

- Differences between c_0 and c_1 become more costly as σ rises

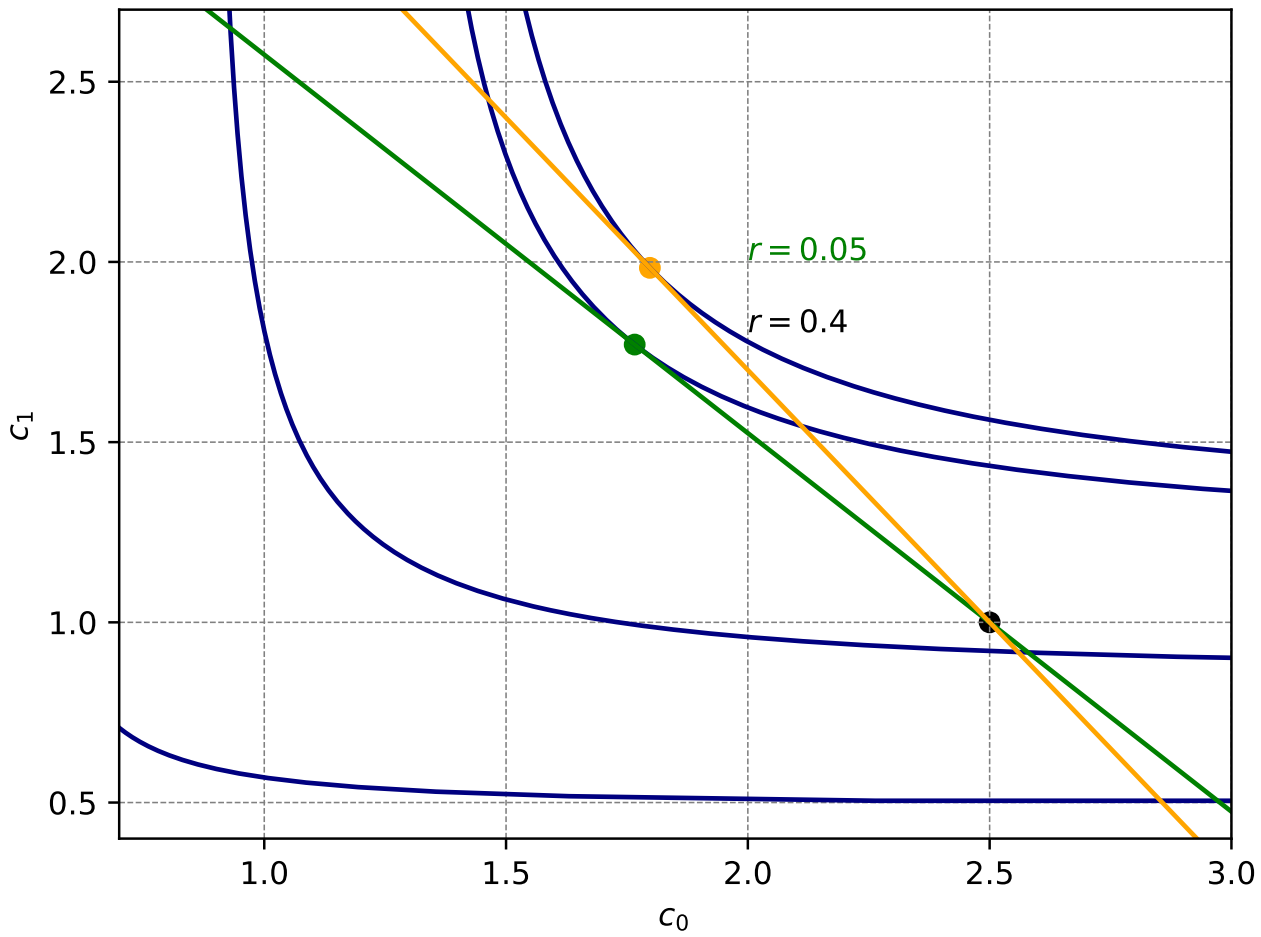


Risk aversion – $u'(c_0) = \beta(1+r)u'(c_1)$



- Risk averse agents smooth more $\Longrightarrow$ stay away from the edges

Interest rates – $u'(c_0) = \beta(1+r)u'(c_1)$



- Higher interest rates lead to more **consumption growth**

Growth vs levels

$$\frac{c_1}{c_0} = [\beta(1+r)]^{\frac{1}{\sigma}}$$

Consumption growth

- An increase in r **always** increases consumption growth
- It's more attractive to consume **relatively** more in the future

Consumption levels

- The increase in consumption growth can be achieved in many ways
  - $c_0 \downarrow, \quad c_1 -$
  - $c_0 -, \quad c_1 \uparrow$
  - $c_0 \uparrow, \quad c_1 \uparrow\uparrow$
  - etc.
- How c_0 and c_1 react to a change in r is not obvious

Income and substitution effects

Remember micro: effect of price changes can be decomposed!

- Interest rate r is the relative price of period 1 vs period 0 consumption

Substitution effect

- Holding utility at the initial level, how does the optimal bundle change when the interest rate changes (i.e., the budget slope changes)?
- This effect becomes strongest with $\sigma \to 0$, as agents consumer all in one of the periods (corner solutions)

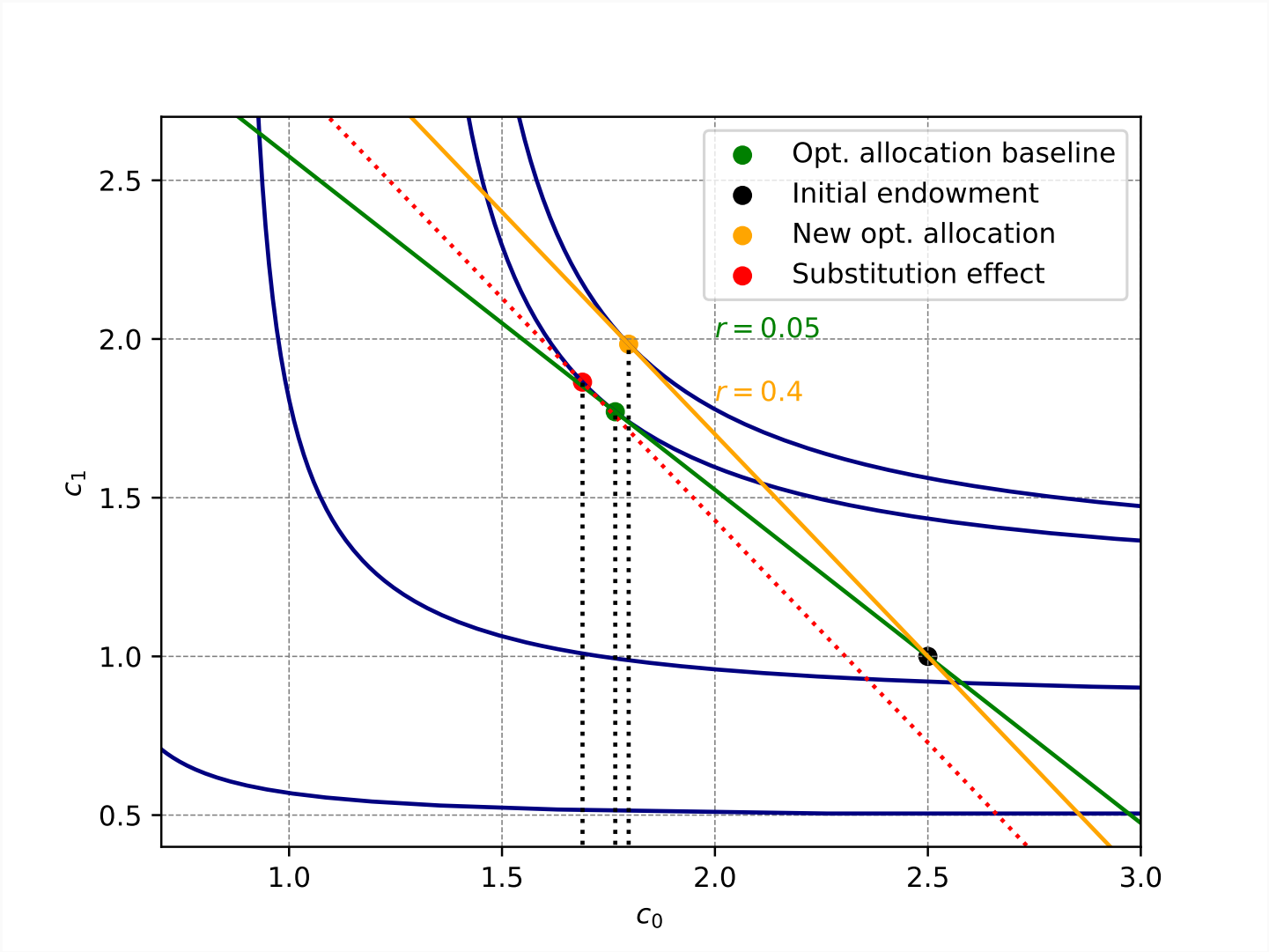
Income effect

- Higher interest rate changes lifetime resources depending on endowment timing

Danger: Wealth effect

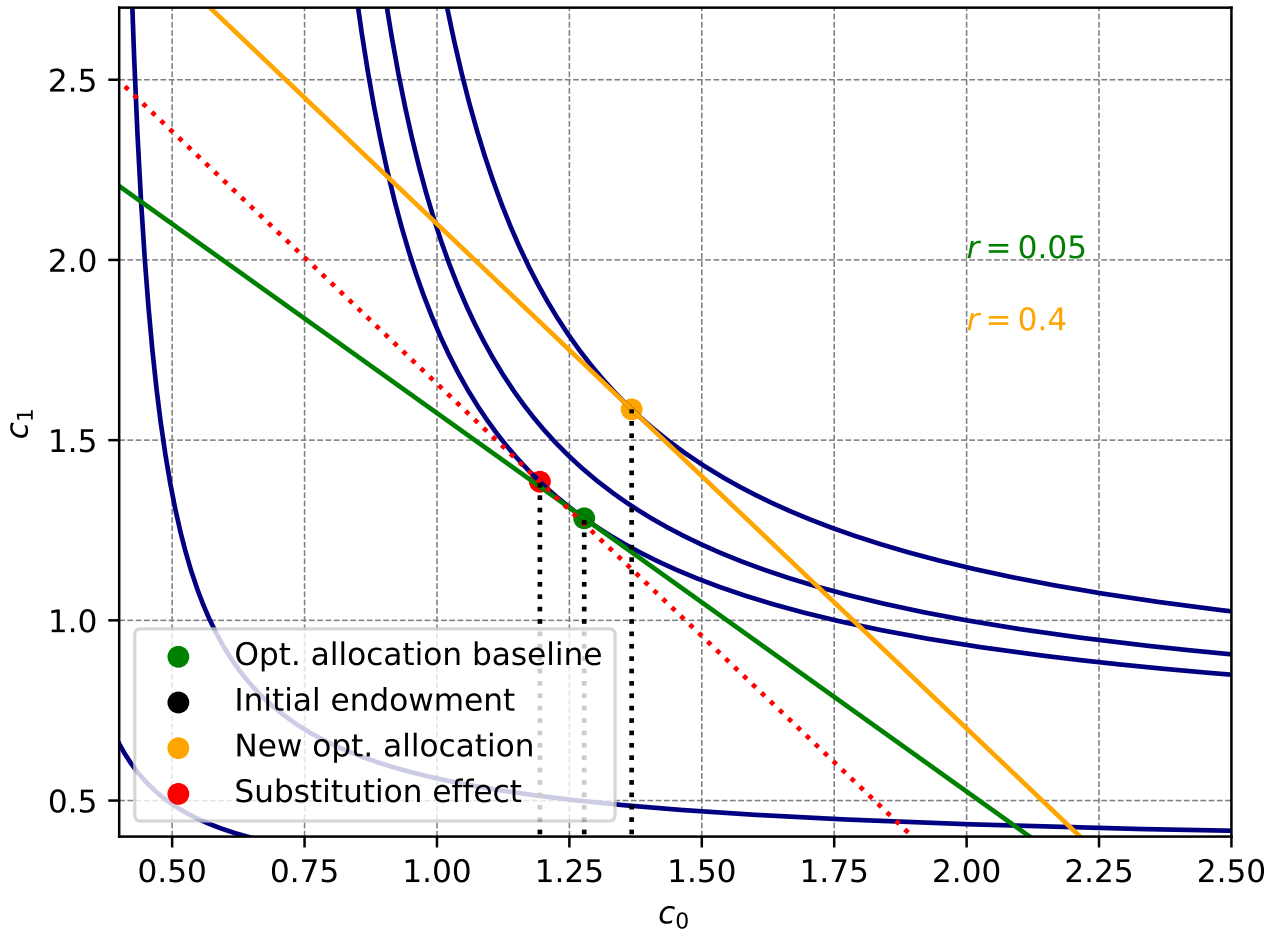
- If r rises, tomorrow's income is worth less today

Interest rates – $u'(c_0) = \beta(1+r)u'(c_1)$



- Substitution, income and wealth effect change choices

Interest rates, $u'(c_0) = \beta(1+r)u'(c_1)$ with $y_1 = 0$



- Substitution effect: $\partial c_0/\partial r < 0$ because of higher return
- Income effect: $\partial c_0/\partial r > 0$ because agent is richer

Income and substitution effect

Period 0 consumption as a function of parameters

$$c_0^{-\sigma} = \beta(1+r)c_1^{-\sigma} \text{ and } c_1 = y_1 + (1+r)(y_0 - c_0)$$

$$\implies c_0\left(1 + \frac{(\beta(1+r))^{\frac{1}{\sigma}}}{1+r}\right) = \frac{y_1}{1+r} + y_0$$

Effects of interest rate changes

- Wealth effect (goes to 0 as $y_1 \to 0$)
- Income effect
- Substitution effect (goes to 0 as $\sigma \to \infty$)

Income and substitution effect

Period 0 consumption as a function of parameters

$$c_0^{-\sigma} = \beta(1+r)c_1^{-\sigma} \text{ and } c_1 = y_1 + (1+r)(y_0 - c_0)$$

$$\implies c_0\left(1 + \frac{(\beta(1+r))^{\frac{1}{\sigma}}}{1+r}\right) = \frac{y_1}{1+r} + y_0$$

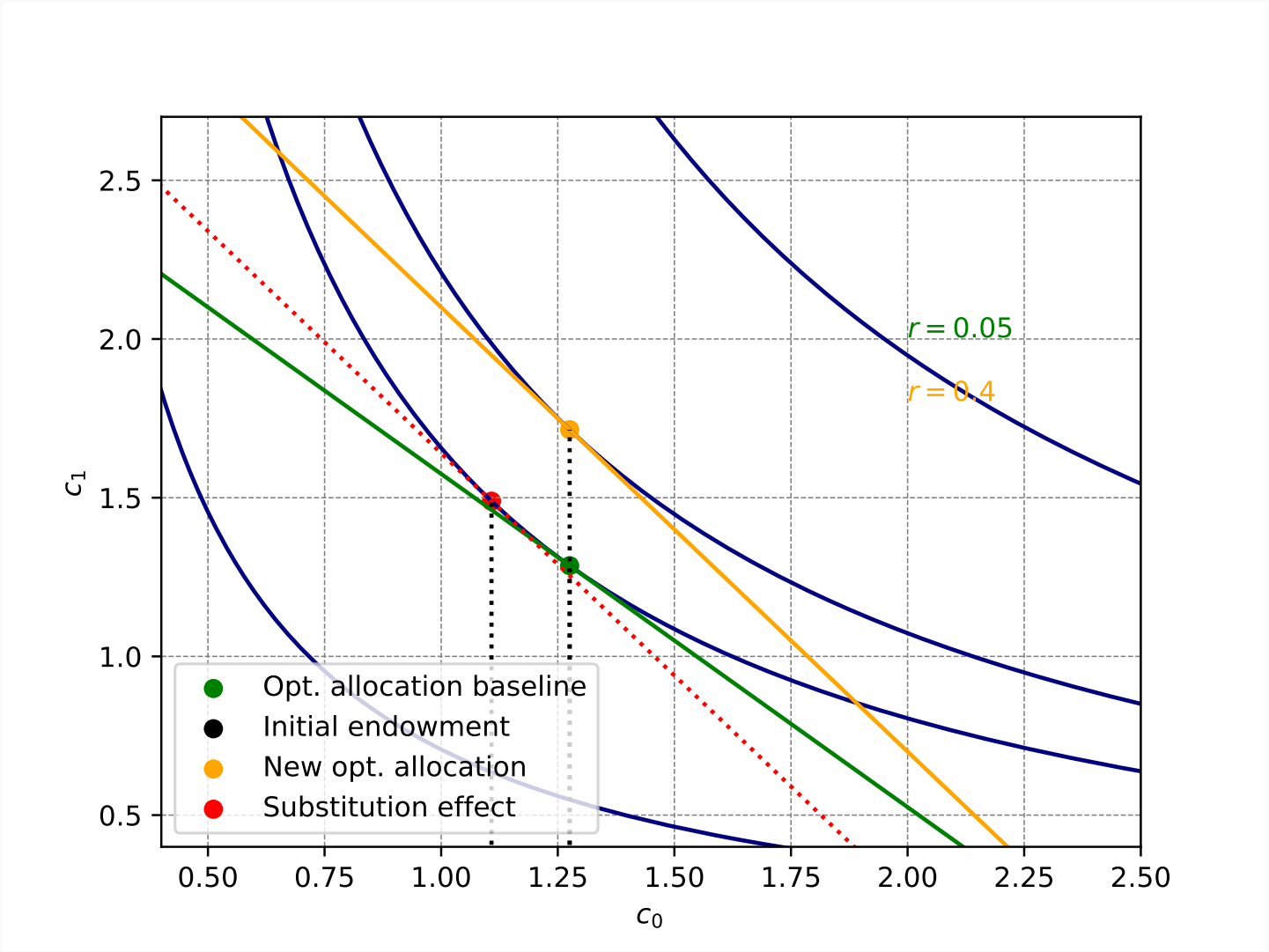
Effects of interest rate changes

- Wealth effect (goes to 0 as $y_1 \to 0$)
- Income effect
- Substitution effect (goes to 0 as $\sigma \to \infty$)

Income and substitution effects cancel if $\sigma = 1$, no wealth effect if $y_1 = 0$

$$c_0\,(1+\beta) = y_0$$

Interest rates – $u'(c_0) = \beta(1+r)u'(c_1)$



- With log-utility and $y_1 = 0$ (no wealth effect), the effects cancel

Appendix

Ramsey model

Frank Ramsey

- English mathematical genius, died early in 1930
- Publishes 'A mathematical theory of saving' in 1928

The model

- Ramsey's theory + Solow model form the neoclassical growth model
- Savings an endogenous object (exogenous in Solow)
- Expectations matter for consumption

Relevance

- Foundation for modern macroeconomics
- What should the role of governments be? What drives growth? What is the role of fiscal policy?

Overlapping generations model

Peter A. Diamond (student of Solow at MIT)

- Nobel Prize winner in 2010 (for something else)

The model

- Different generations modelled explicitly (heterogeneity)

Relevance

- Social security and pension system design
- Design of educational systems
- Financing of public debt

The RBC model

Finn Kydland (NO) & Edward Prescott (US) (Carnegie Mellon)

- Nobel Prize winners in 2004

The model

- Neo-classical growth model with short-run disturbances
- Business cycles are the result of supply shocks

Relevance

- Economic stabilization policies

The DMP model

Peter Diamond, Dale Mortensen, and Christopher Pissarides

- Nobel Prize winners in 2010

The model

- Moving away from perfectly competitive labor markets
- Search frictions result in involuntary unemployment and wage inequality

Relevance

- Design of unemployment insurance schemes

Nominal rigidities and the New Keynesian model

New Keynesian Economics

- No Nobel Prize winners yet

The model

- Moving away from perfect towards monopolistic competition
- Incorporating rigid prices into the RBC model
- Money is no longer neutral, monetary policy matters

Relevance

- Monetary policy
- Interest rates
- Inflation

Monetary policy making

Torsten Persson (SE) & Guido Tabellini (IT)

- Not Nobel Prize winners yet

The model

- Game theoretical framework of society
- Expectations, reputation and credibility matter
- Allows prescriptions on optimal policy

Relevance

- Institutional design of monetary policy institutions